



Putting **Firepower** into the Next Generation Firewall

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CCIE #2926

January 30, 2018

Cisco
Connect

Your Time
Is Now





Today's agenda

- Firepower Software Overview
- ASA and Firepower NGFW platforms
- Management options
- New Capabilities in 6.2.2
- Deployment Design Use-case
- Policies

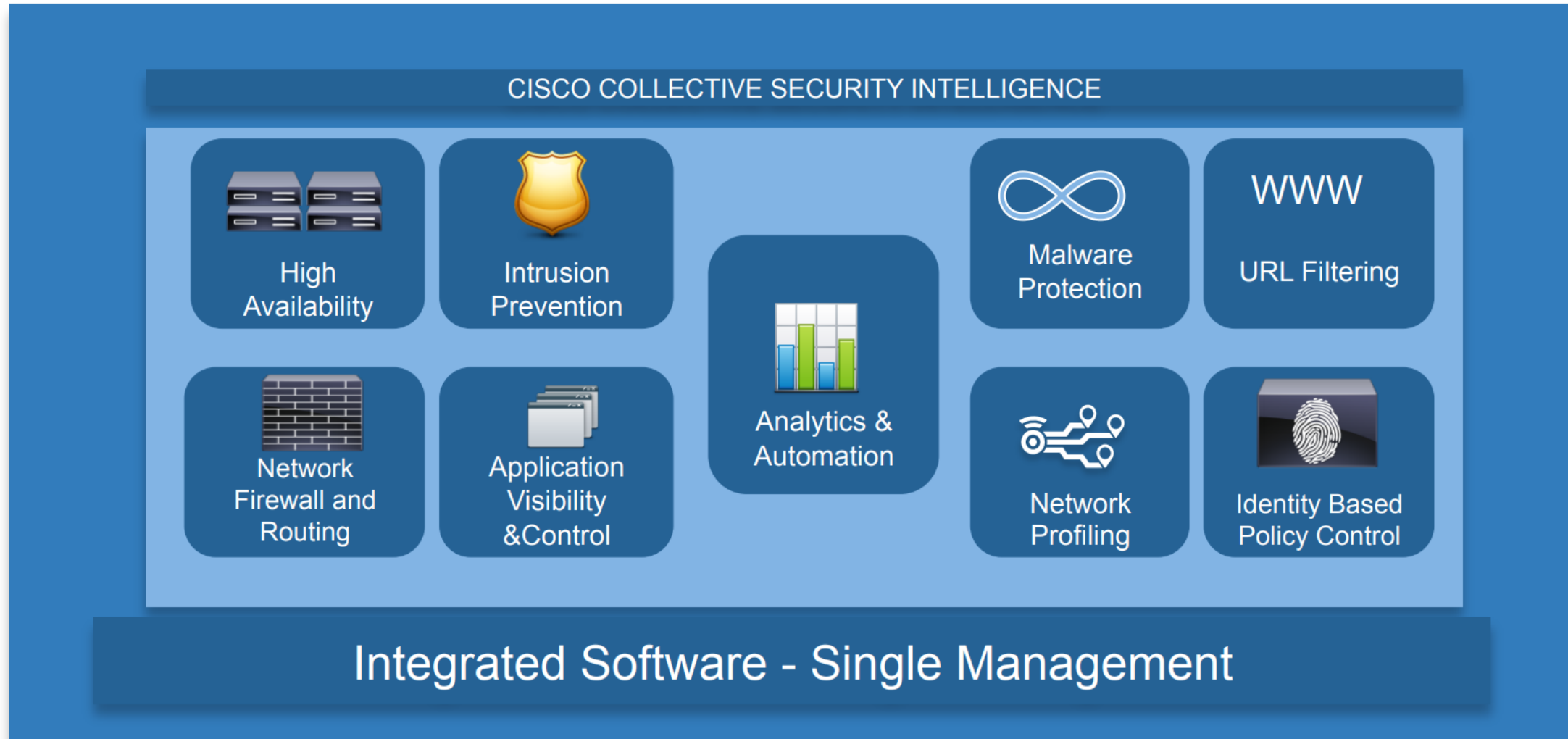
Cisco
Connect



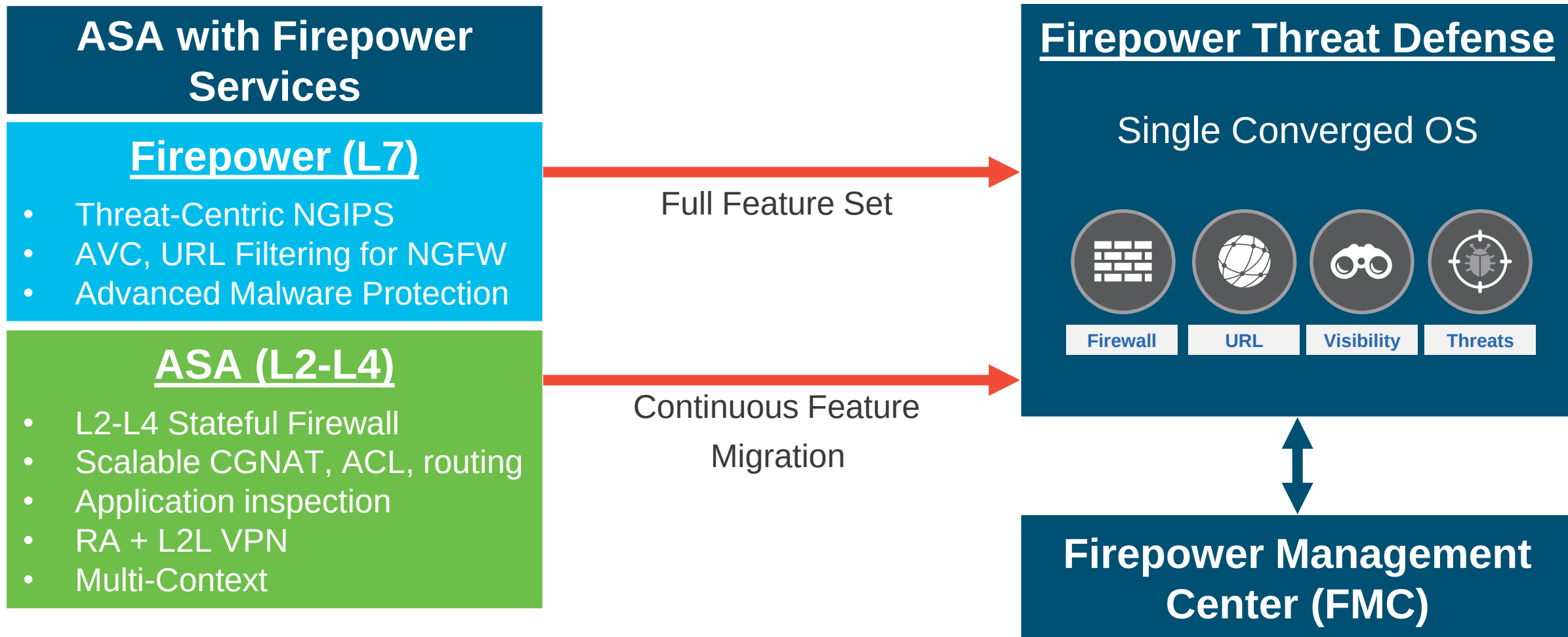
Firepower NGFW Software



Firepower Threat Defense (FTD)



Firepower Threat Defense



What are the Firepower Deployment Options?

Firepower Appliances



7000/7100/8000/Virtual



ASA with Firepower Services



ASA 5500X (all models)



Firepower Threat Defense



ASA 5500X / Virtual
Firepower 2100 / 4100 / 9300



5585 cannot run FTD Image!

All Managed by Firepower Management Center

Feature Comparison: ASA with Firepower Services and Firepower Threat Defense

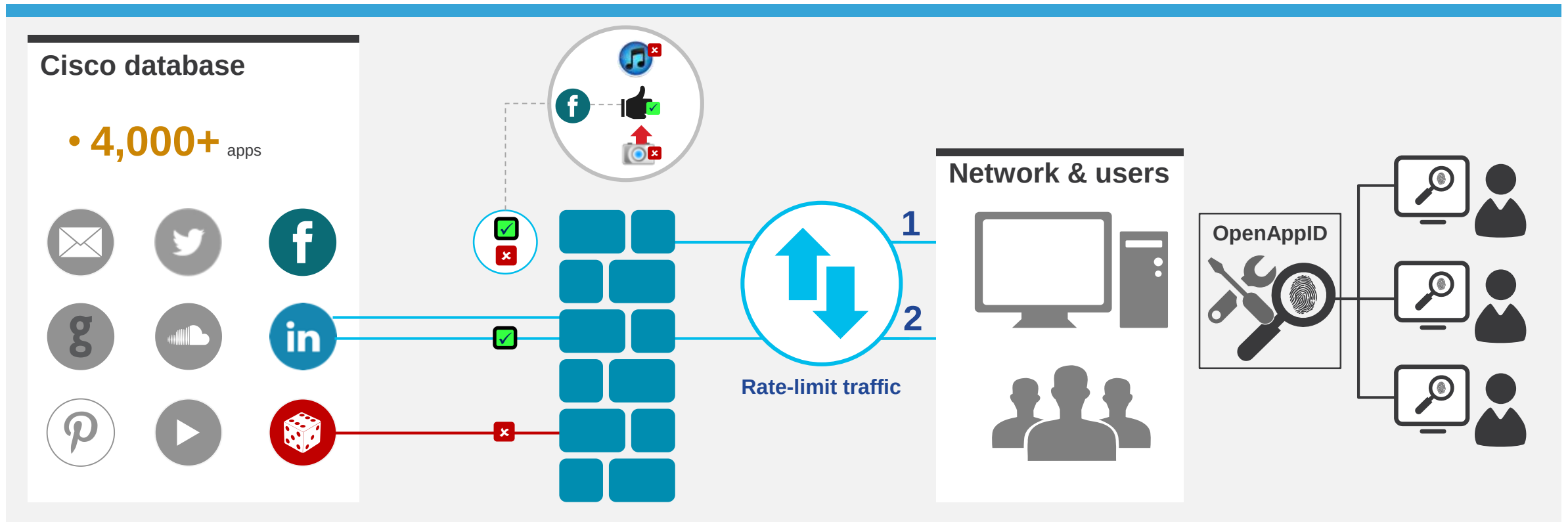
Note: Not an exhaustive feature list

	Features	Firepower Threat Defense	Firepower Services for ASA
SIMILARITIES	Routing +NAT	✓ (OSPF, BGP, Static, RIP, Multicast, EIGRP/PBR via FlexConfig)	✓ (OSPF, BGP, EIGRP, static, RIP, Multicast)
	OnBox Management	✓	✓
	HA (Active/Passive)	✓	✓
	Clustering (Active/Active)	✓	✓
	Site to Site and Remote Access VPN	✓	✓
	Policy based on SGT tags	✓	✓
DIFFERENCES	Unified ASA and Firepower rules and objects	✓	✗
	Hypervisor Support	✓ (AWS, VMware, KVM, Azure 6.2)	✗
	Smart Licensing Support	✓	✗
	Multi-Context Support	✗ (Coming Soon!)	✓



Next-generation visibility with OpenAppID

Application Visibility & Control



See and understand risks

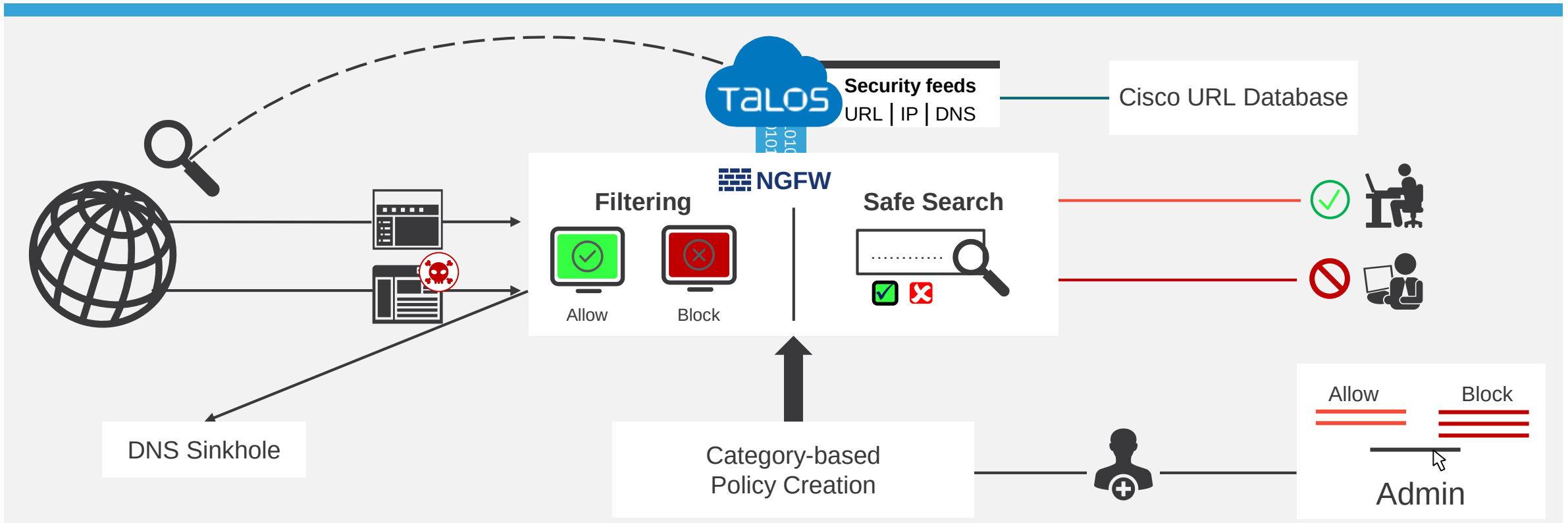
Enforce granular access control

Prioritize traffic and limit rates

Create detectors for custom apps

Web acceptable use controls and threat prevention

URL Filtering – Security Intelligence Feeds – DNS Sinkhole capability



Classify **280M+** URLs

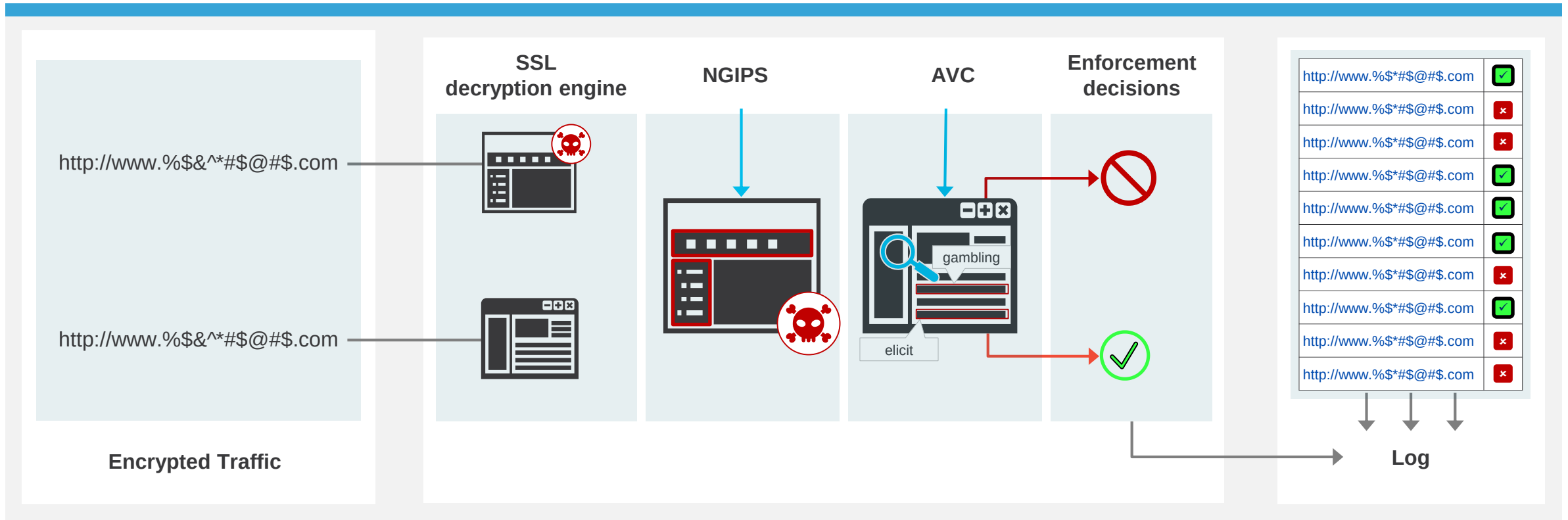
Filter sites using **80+** categories

Manage "allow/block" lists easily

Block latest malicious URLs

Granular SSL Decryption Capabilities

SSL TLS handshake certificate inspection and TLS decryption engine



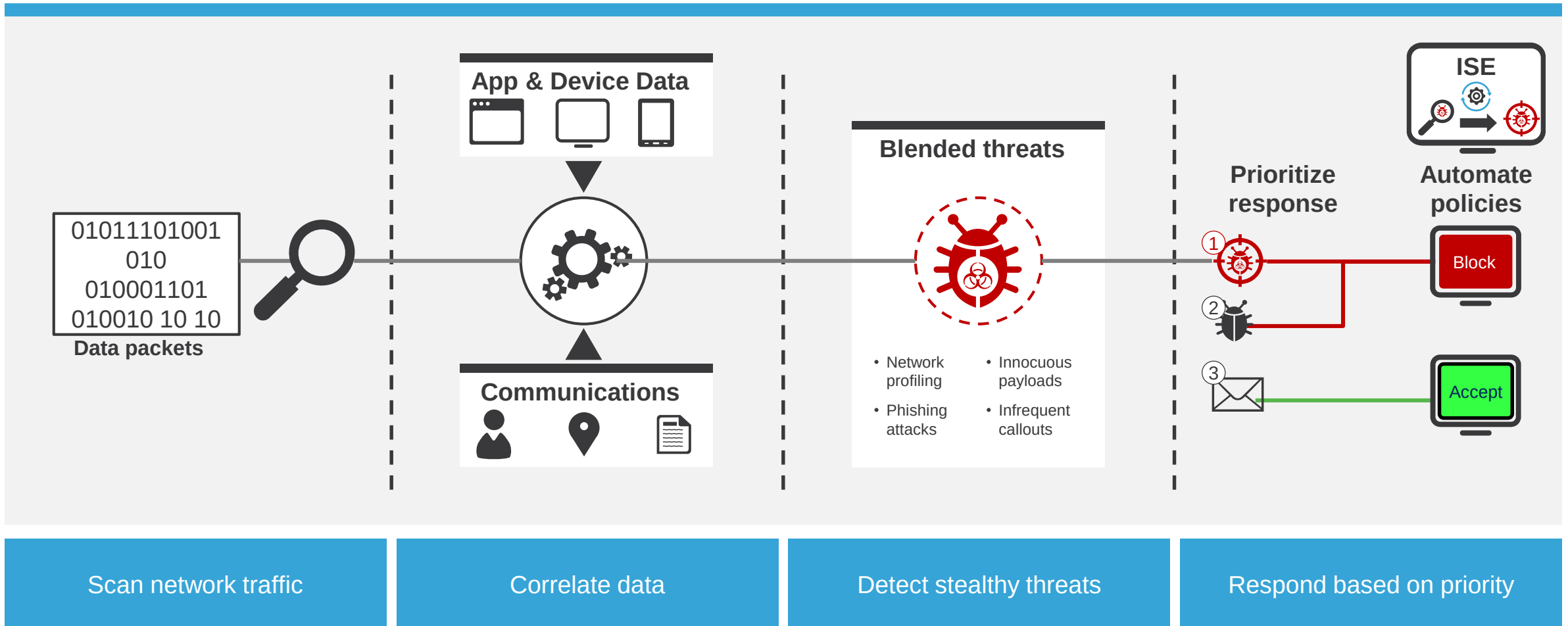
Decrypt **3.5 Gbps** traffic over
five million simultaneous flows

Inspect deciphered packets

Track and log all SSL sessions

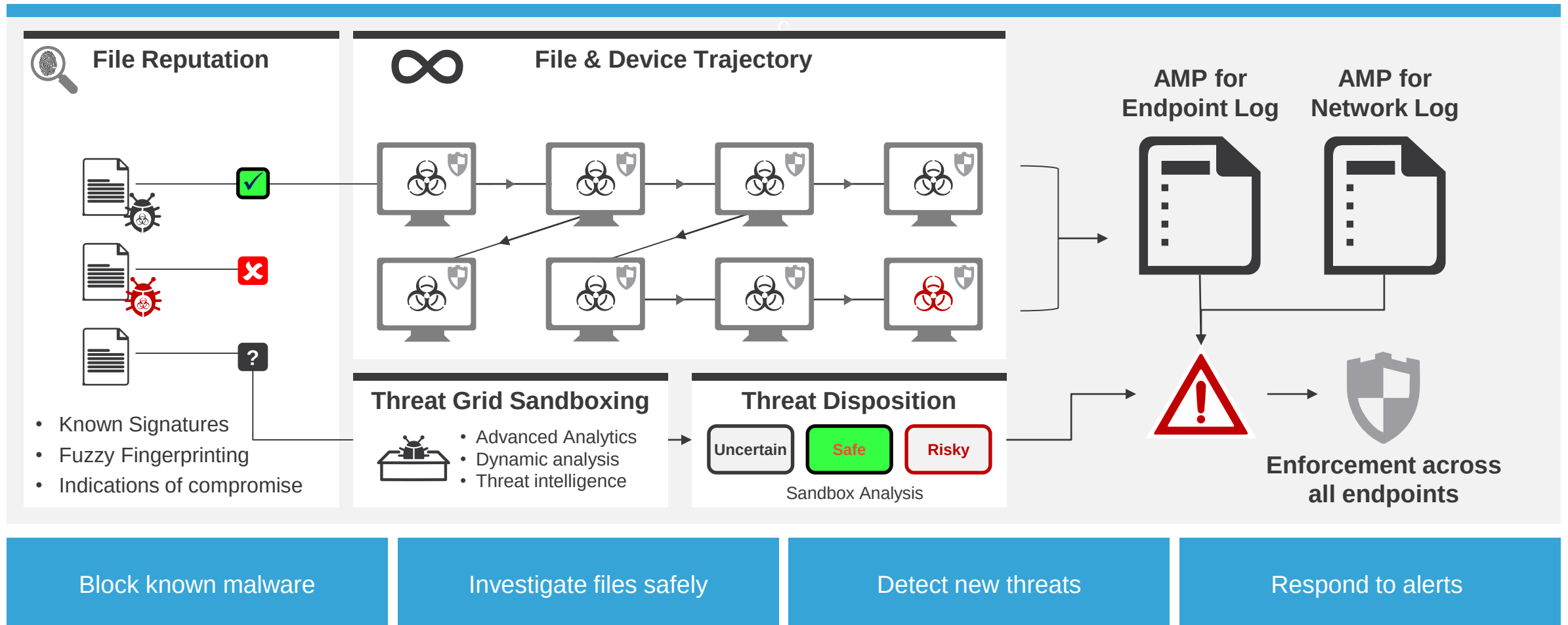
Application and Context aware Intrusion Prevention

Next-Generation Intrusion Prevention System (NGIPS)



Malware and ransomware detection and blocking

Cisco AMP Threat Grid (Advanced Malware Protection and cloud sandboxing)



FlexConfig

- Provides a way to configure ASA features not exposed directly by Firepower Management Center

- | | |
|--|--|
| <ul style="list-style-type: none">• EIGRP Routing• PBR• ISIS Routing• NetFlow (NSEL) export• VXLAN | <ul style="list-style-type: none">• ALG inspections• IPv6 header inspection• Platform Sysopt commands• WCCP |
|--|--|

ASA & Firepower Platforms





Cisco ASA 5500-X

SMB and Enterprise Branch NGFW



5506 / 5508 / 5516
Performance

- 1-Gbp interfaces
- Up to 450 Mbps throughput
- Wireless Option for 5506-X
- Software Switching capability
- Firepower Threat Defense or ASA Software Options



5525 / 5545 / 5555
Performance

- 1-Gbp interfaces
- Up to 1.2 Gbps throughput
- 5545 / 5555 Redundant Power Supply and SSD option
- Firepower Threat Defense or ASA Software Options



Unified Management

- Firepower Management Center (Enterprise Management)
- Firepower Device Manager (On Box Manager)
- Cisco Defense Orchestrator (Cloud Management)

Cisco Firepower 2100 Series

Introducing four high-performance models



Cisco Firepower 2110/2120 Model



Cisco Firepower 2130/2140 Model



Performance and Density Optimization

- 1-Gbp and 10-Gbps interfaces
- Up to 8.5-Gbps throughput
- 1-rack-unit (RU) form factor
- Dual SSD slots
- 12x RJ45 ports, 4xSFP(+)
- 2130 / 2140 Models
 - 1x Network Module
 - Fail to Wire Option*
 - DC & Dual PSU support



Purpose Built NGFW

- Integrated inspection engines for FW, NGIPS, Application Visibility and Control (AVC), URL, Cisco Advanced Malware Protection (AMP)



Unified Management

- Firepower Management Center (Enterprise Management)
- Firepower Device Manager (On Box Manager)
- Cisco Defense Orchestrator (Cloud Management)

Firepower 2100 Series Performance

	FPR 2110	FPR 2120	FPR 2130	FPR 2140
Throughput NGFW	1.9 Gbps	3 Gbps	4.75 Gbps	8.5 Gbps
Throughput NGFW + IPS	1.9 Gbps	3 Gbps	4.75 Gbps	8.5 Gbps
Maximum concurrent sessions	1 M	1.2 M	2 M	3.5 M
Maximum new connections per second	12000	16000	24000	40000

NO DROP IN PERFORMANCE!

Cisco Firepower 4100 Series

High performance campus and data center



Performance and Density Optimization

- 10-Gb and 40-Gb interfaces
- Up to 24-Gbps throughput
- 1-rack-unit (RU) form factor
- Low latency



Multiservice Security

- Integrated inspection engines for FW, NGIPS, Application Visibility and Control (AVC), URL, Cisco Advanced Malware Protection (AMP)
- Radware DefensePro DDoS and other future third party



Unified Management

- Firepower Management Center (Enterprise Management)
- **Firepower Device Manager (On Box Manager)**
- Cisco Defense Orchestrator (Cloud Management)



Cisco Firepower 9300 Platform

High performance data center



Modular

Benefits

- Standards and interoperability
- Flexible architecture

Features

- Template-driven security
- Secure containerization for customer apps
- RESTful/JSON API
- Third-party orchestration and management



Multiservice
Security

Benefits

- Integration of best-in-class security
- Dynamic service stitching

Features

- ASA container option
- Firepower™ Threat Defense:
 - NGIPS, AMP, URL, AVC
- Third-party containers:
 - Radware DDoS



Carrier Class

Features

- Compact, 3RU form factor
- 10-Gbps/40-Gbps I/O; 100-Gbps
- Terabit backplane
- Low latency, intelligent fast path
- Network Equipment-Building System (NEBS) ready

Cisco NGFW Platforms

Up to 6x with clustering!



Firepower Threat Defense for
ASA 5500-X



250 Mb -> 1.75 Gb
(NGFW + IPS Throughput)

Firepower 2100 Series



2 Gb -> 8 GB
(NGFW + IPS Throughput)

Firepower 4100 Series
and Firepower 9300



41xx = 10 Gb -> 24 Gb
93xx = 24 Gb -> 53Gb

NGFW capabilities all managed by Firepower Management Center

Software Support – Physical Platforms

	ASA	Firepower NGIPS	ASA with FirePOWER Services	Firepower Threat Defense
ASA 5506X -> 5555X (all models)	✓		✓	✓
Firepower 2100 (all models)	✓			✓
Firepower 4100 (all models)	✓			✓
Firepower 9300 (all models)	✓			✓
ASA 5585 (With SSP blade)	✓		✓	
Firepower 7000 / 8000 (IPS appliances)		✓		

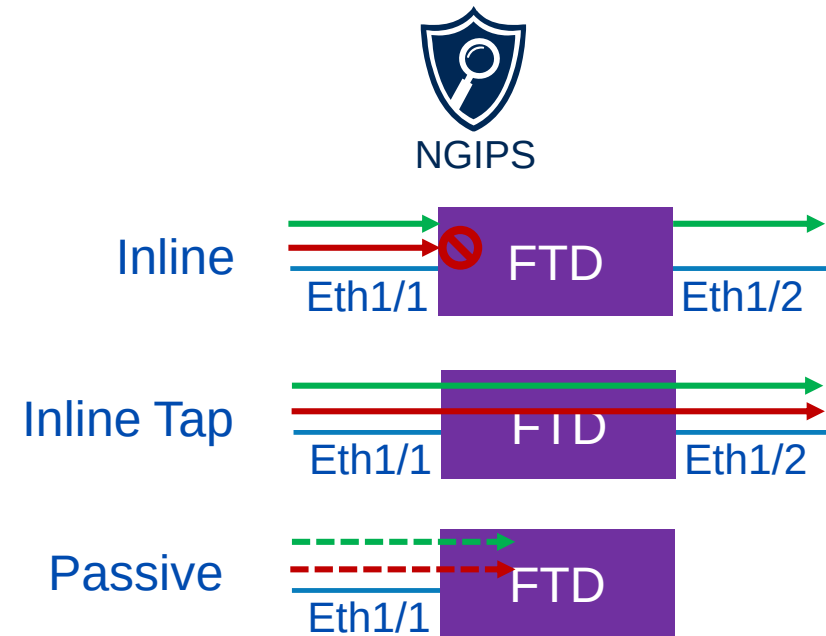
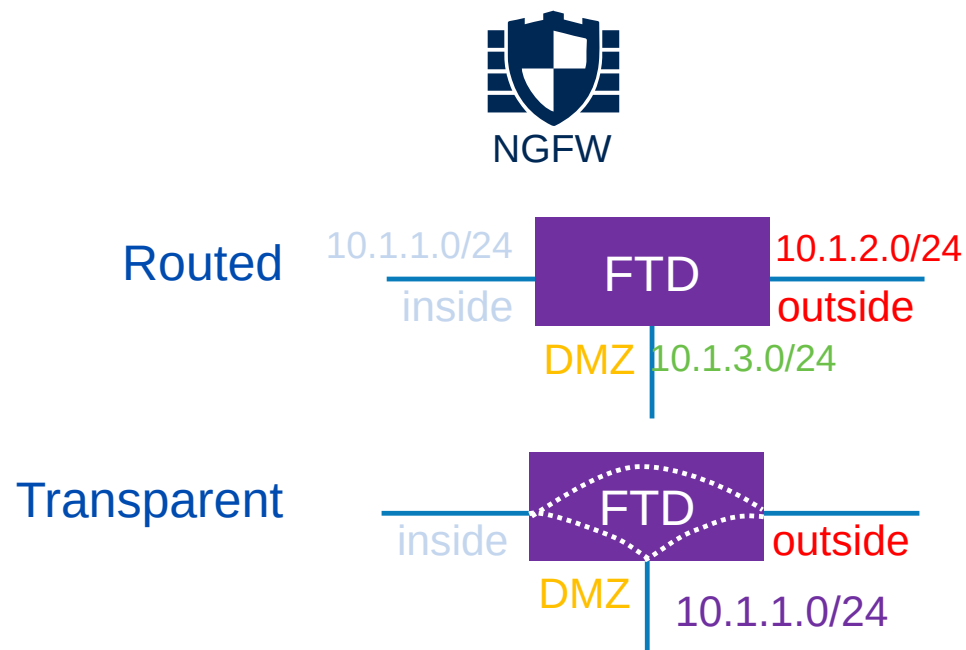


Software Support - Virtual Platforms

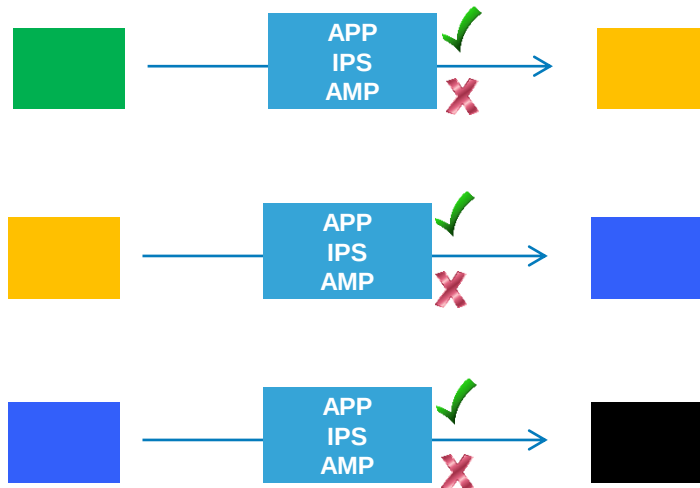
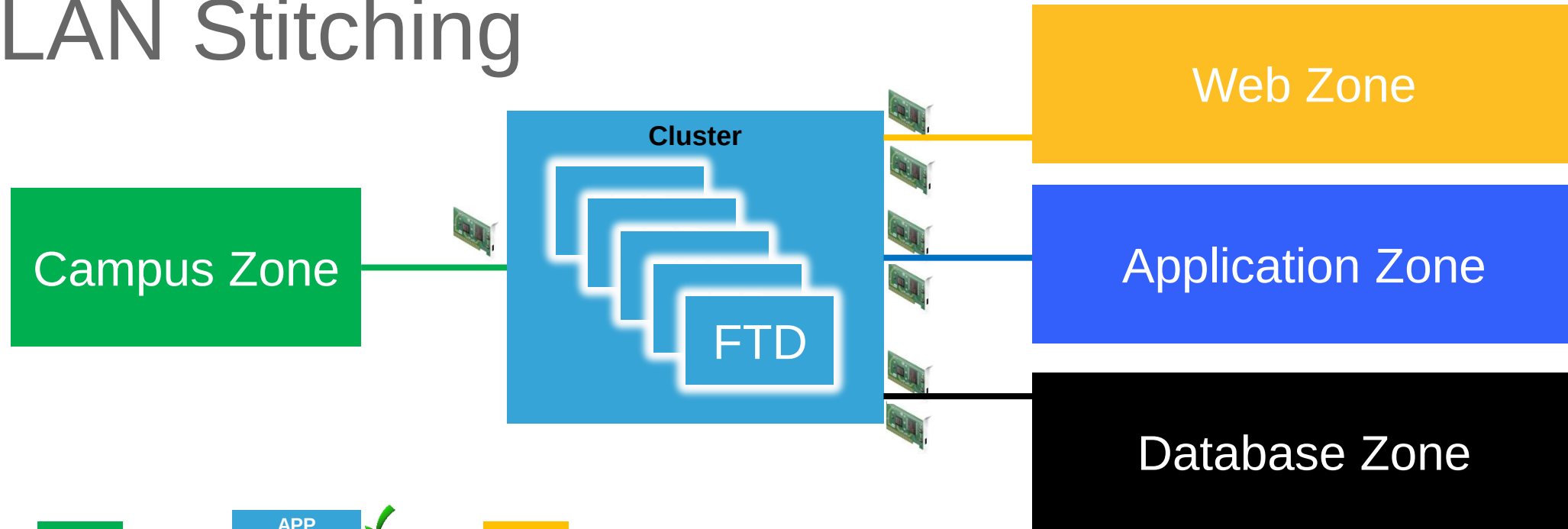
	ASA	Firepower NGIPS	Firepower Threat Defense
ASAv (vSphere, AWS, Azure, Hyper-V, KVM)	✓		
Firepower NGIPSv (vSphere + ISR UCS-E)		✓	
Firepower NGFWv (vSphere, AWS, Azure, KVM)			✓

FTD Deployment Modes

- **FTD** can act as both NGFW and NGIPS on different network interfaces
NGIPS operates as standalone Firepower with limited ASA data plane functionality

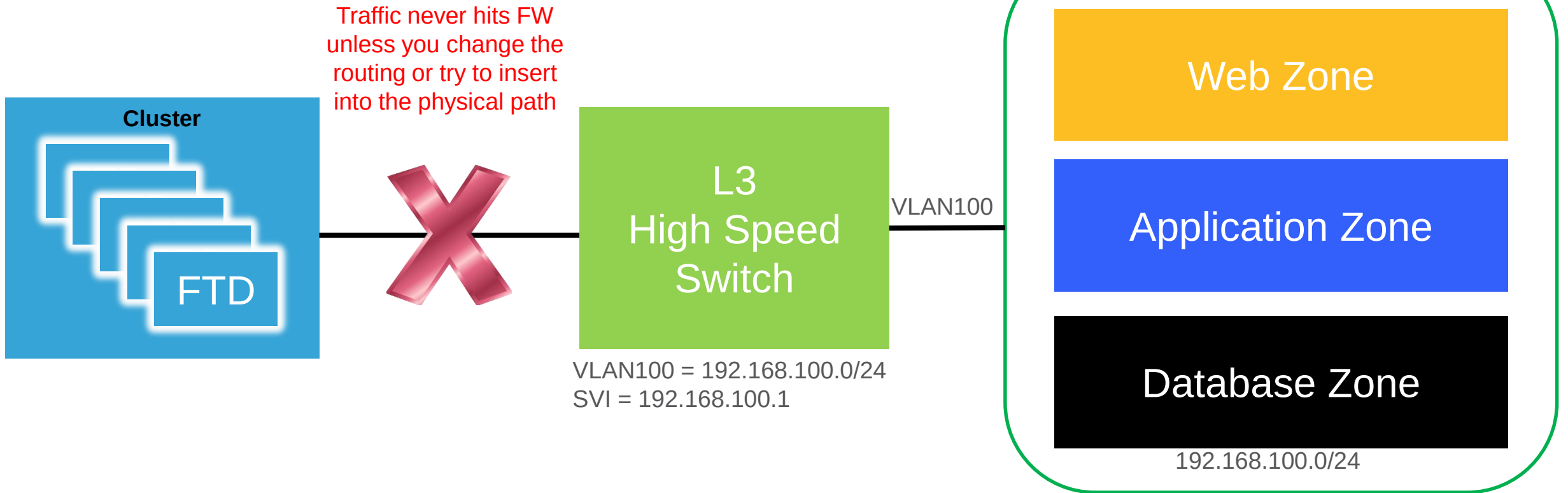


VLAN Stitching



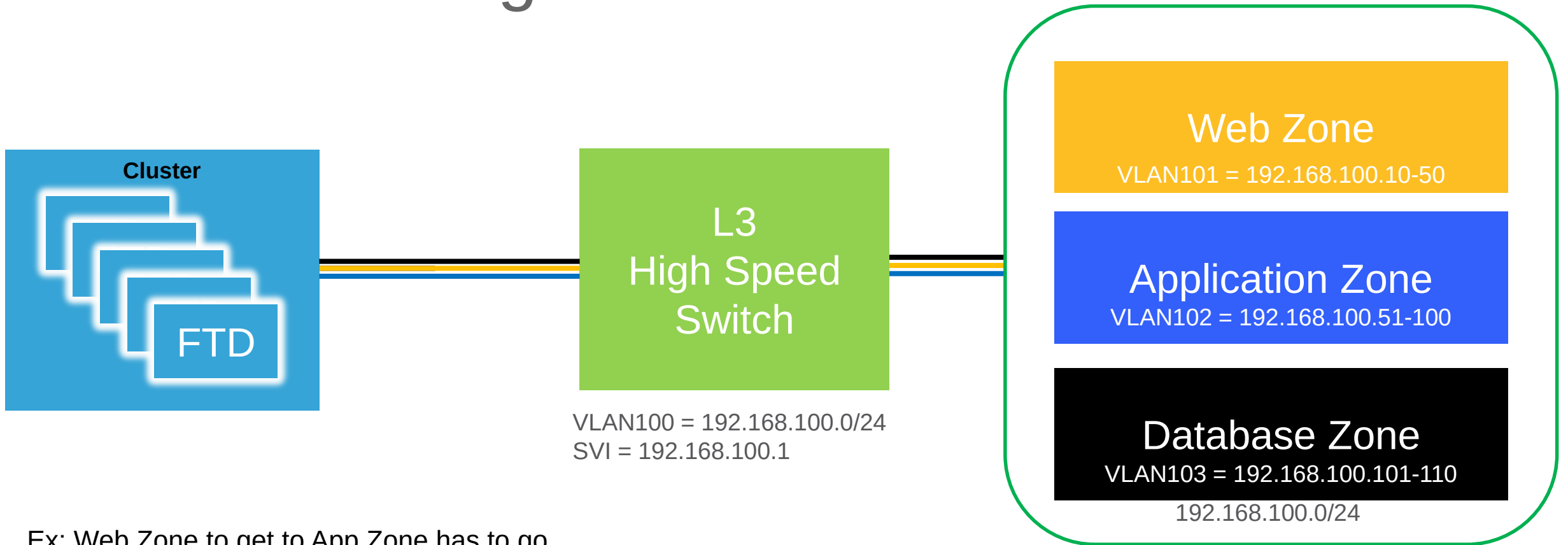
How do I insert this into the Datacenter without having to change the physical infrastructure or move the routing?

VLAN Stitching - Before



How do I insert this into the Datacenter without having to change the physical infrastructure or move the routing?

VLAN Stitching - After



Ex: Web Zone to get to App Zone has to go through policy on FTD. FTD stitches VLAN 101, 102 and 103. Now I can add additional L7 Inspection.

How do I insert this into the Datacenter without having to change the physical infrastructure or move the routing?

The Security-Performance Problem



Bypass Options

Fail-to-Wire Interfaces	Bypass traffic upon appliance failure, including loss of power.
Automatic Application Bypass	Restarts Snort processes upon degraded performance
Intelligent Application Bypass	Application-specific acceleration of defined applications if performance is degraded
Trust Rules	Acceleration defined traffic but still apply Security Intelligence
Prefilter Policy	Bypass deep inspection and Security Intelligence based on Port / Protocol / IP Address / Zone

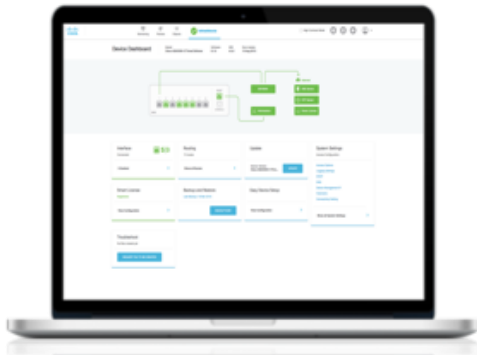
Management Platform Options



Management Options

On-box

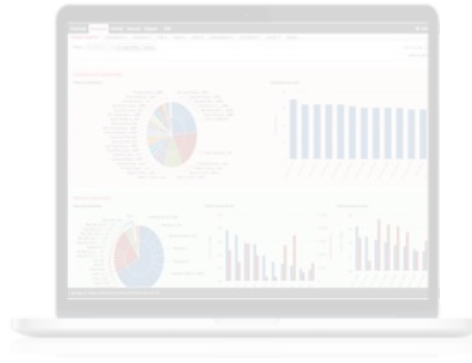
Firepower Device Manager



Enables easy on-box management of common security and policy tasks

Centralized

Firepower Management Center



Enables comprehensive security administration and automation of multiple appliances

Cloud-based

Cisco Defense Orchestrator

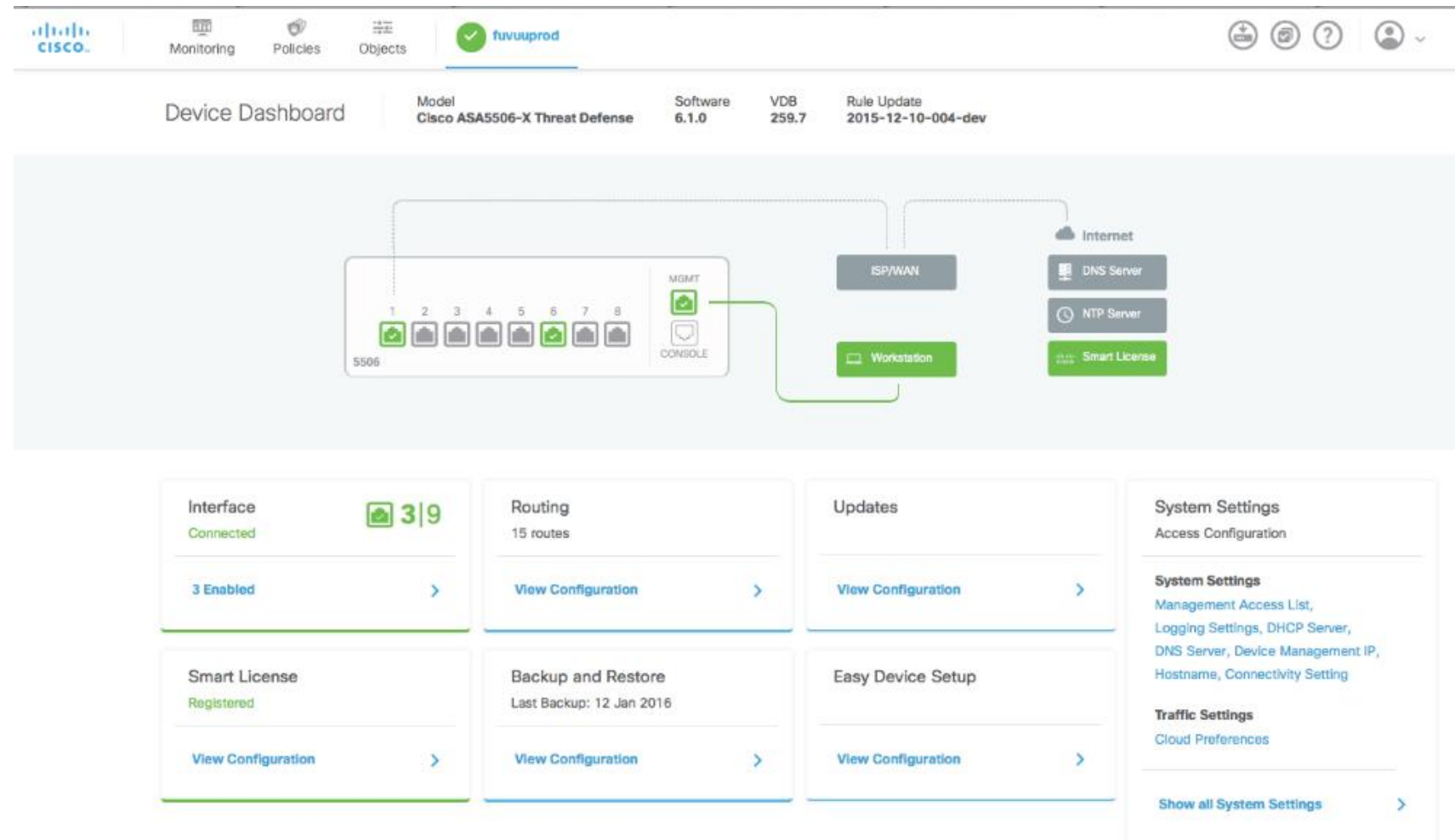


Enables centralized cloud-based policy management of multiple deployments



Firepower Device Manager

- On-box manager for managing a single Firepower Threat Defense device
- Targeted for SMB market
- Designed for Networking Security Administrator
- Simple & Intuitive
- Mutually Exclusive from FMC
- CLI for troubleshooting



Management Options

On-box



Enables easy on-box management of common security and policy tasks

Centralized

Firepower Management Center



Enables comprehensive security administration and automation of multiple appliances

Cloud-based



Enables centralized cloud-based policy management of multiple deployments

Firepower Management Center

- Single manager for Firepower Threat Defense
- Can also manage Firepower appliance and “Services” deployments
- Broadest set of security capabilities for Firepower platforms!

Rules																
Security Intelligence HTTP Responses Advanced																
Filter by Device										Add Category		Add Rule		Search Rules		
#	Name	Source Zones	Dest Zones	Source Networks	Dest Networks	VLAN Tags	Users	Applications	Src Ports	Dest Ports	URLs	Action				
Mandatory - acl (1-5)																
1	test-rule	any	any	1.1.1.1 2.2.2.2	1.1.1.1 2.2.2.2	any	any	any	any	HTTPS	any	Block				0
2	r2	any	any	IPv4-Private-1	any	any	any	any	any	any	any	Block				0

Management Options

On-box



Enables easy on-box management of common security and policy tasks

Centralized

Firepower Management Center



Enables comprehensive security administration and automation of multiple appliances

Cloud-based



Cisco Defense Orchestrator



Enables centralized cloud-based policy management of multiple deployments

cisco On-box vs Off-box



	Firepower Management Center (Off-box)	Firepower Device Manager (On-box)
NAT & Routing	✓	✓
Access Control	✓	✓
Intrusion & Malware	✓	✓
Device & Events Monitoring	✓	✓
VPN - Site to Site & RA	✓	✓
Security Intelligence	✓	✗
Other Policies: SSL, Identity, Rate Limiting (QoS) etc.	✓	✗
Active/Passive Authentications	✓	✓
Firewall Mode	Router / Transparent	Routed
Threat Intelligence & Analytics	✓	✗
Correlation & Remediation	✓	✗
Risk Reports	✓	✗
Device Setup Wizard	✗	✓
IPS Tuning	✓	✗
High Availability	✓	✗

Firepower Management Center New Capabilities





Troubleshooting: Packet Tracer

- Displays logs for a single simulated (virtual) packet
- Tracing data **will include information from Snort & preprocessors** about verdicts and actions taken while processing a packet

Overview Analysis Policies Devices Objects AMP Deploy System Help admin

Configuration Users Domains Integration Updates Licenses Health Monitor Monitoring Tools

Advanced Troubleshooting

192.168.0.165

File Download Threat Defense CLI **Packet Tracer** Capture w/Trace

Select the packet type and supply the packet parameters. Click start to trace the packet.

Packet type:	TCP	Interface*:	b2
Source*:	IP address (IPv4) 12.1.11.1	Source Port*:	12
Destination*:	IP address (IPv4) 1.1.11.1	Destination Port*:	http
SGT number:	SGT number. (0-65533)	VLAN ID:	VLAN ID... (1-4096)
Output Format:	summary	Destination Mac Address:	XXXX.XXXX.XXXX

Start Clear

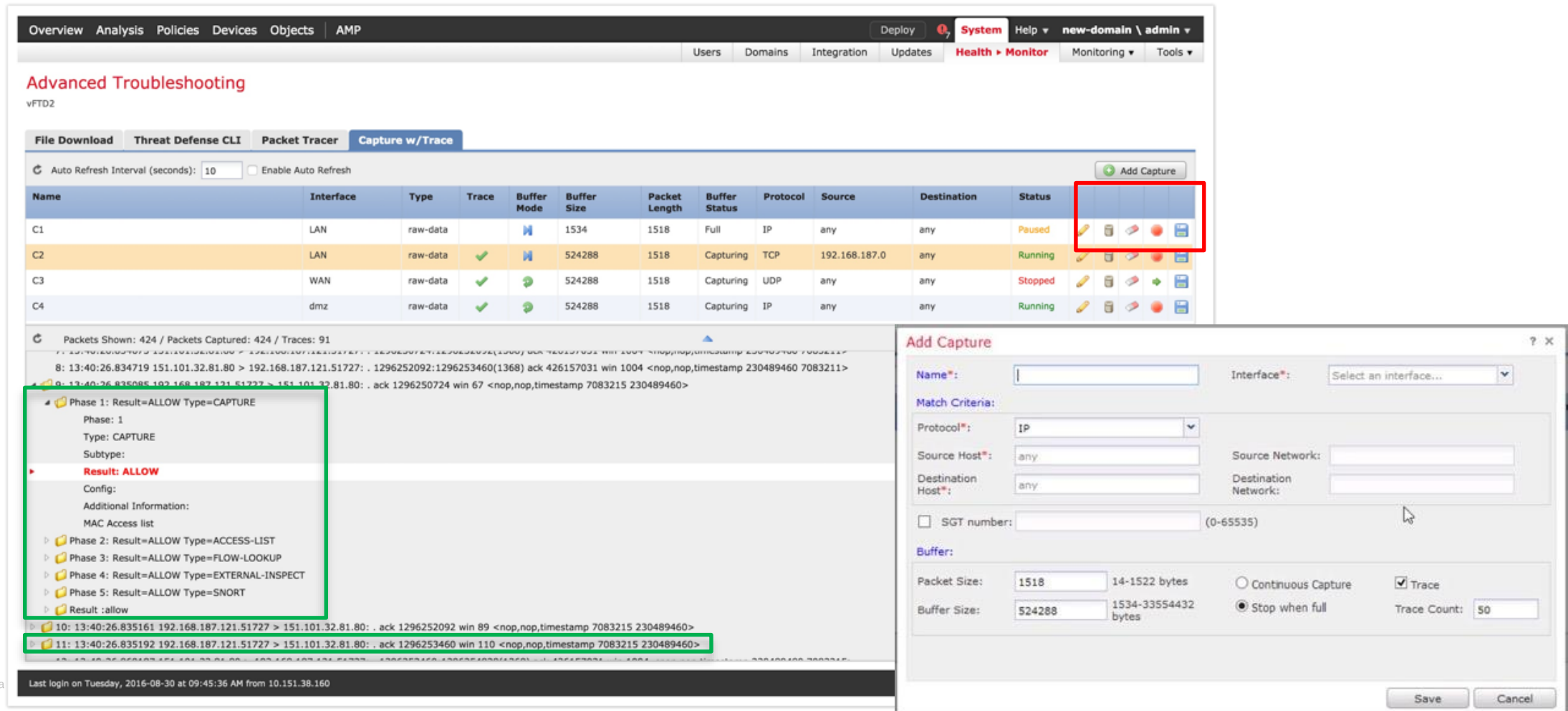
Output Tree

- Phase 1: Result=ALLOW Type=FLOW-LOOKUP
Type: FLOW-LOOKUP
Subtype:
Result: ALLOW
Config:
Additional Information:
Found flow with id 589, using existing flow
- Phase 2: Result=ALLOW Type=EXTERNAL-INSPECT
- Phase 3: Result=DROP Type=SNORT
Result: drop

Last login on Monday, 2016-08-29 at 22:35:11 PM from 10.82.234.205

Troubleshooting: Packet Capture with Trace

- Captures and displays packets from live traffic
- Allows PCAP file download of the capture buffer



The screenshot displays the Cisco Advanced Troubleshooting interface, specifically the 'Capture w/Trace' tab. The interface shows a table of active captures and a detailed view of the selected capture (C2).

Name	Interface	Type	Trace	Buffer Mode	Buffer Size	Packet Length	Buffer Status	Protocol	Source	Destination	Status
C1	LAN	raw-data			1534	1518	Full	IP	any	any	Paused
C2	LAN	raw-data	✓		524288	1518	Capturing	TCP	192.168.187.0	any	Running
C3	WAN	raw-data	✓		524288	1518	Capturing	UDP	any	any	Stopped
C4	dmz	raw-data	✓		524288	1518	Capturing	IP	any	any	Running

The 'Add Capture' dialog box is open, showing the following configuration:

- Name:** (empty)
- Interface:** Select an interface...
- Match Criteria:**
 - Protocol:** IP
 - Source Host:** any
 - Destination Host:** any
 - Source Network:** (empty)
 - Destination Network:** (empty)
 - SGT number:** (empty) (0-65535)
- Buffer:**
 - Packet Size:** 1518 (14-1522 bytes)
 - Buffer Size:** 524288 (1534-33554432 bytes)
 - Continuous Capture:** ☐
 - Stop when full:** ☒
 - Trace:** ☒
 - Trace Count:** 50

The packet capture results for C2 are displayed below the table, showing a list of captured packets. The first packet is highlighted:

```

10: 13:40:26.835161 192.168.187.121.51727 > 151.101.32.81.80: . ack 1296252092 win 89 <nop,nop,timestamp 7083215 230489460>
11: 13:40:26.835192 192.168.187.121.51727 > 151.101.32.81.80: . ack 1296253460 win 110 <nop,nop,timestamp 7083215 230489460>
  
```

The interface also shows a detailed view of the selected packet (11), including the phase and result of the capture.



Lookup features – Geolocation & WHOIS

The screenshot shows the Cisco Firepower Management Center (FMC) interface. The top navigation bar includes tabs for Overview, Analysis (selected), Policies, Devices, Objects, and AMP. Below this is a sub-navigation bar with Context Explorer, Connections, Intrusions, Files, Hosts, Users, Vulnerabilities, Correlation, Custom, Lookup (selected), and Search. The Lookup dropdown menu is open, showing Geolocation and Whois options. The main content area displays a search box with the text "IP Addresses: (Limit 250)" and a list of IP addresses: 72.163.4.161 and 161.148.175.40. Below the search box are Search and Clear buttons. The bottom of the interface shows a table with the results of the lookup.

IP Address	Country	Country Code	Continent
161.148.175.40	Brazil	76	South America
72.163.4.161	United States	840	North America



Lookup Feature: URL

Overview **Analysis** Policies Devices Objects AMP Deploy System Help admin

Context Explorer Connections Intrusions Files Hosts Users Vulnerabilities Correlation Custom **Lookup URL**

Geolocation

URL

Whois

URLs:
(Limit 250)

cisco.com
developmentserver.com
google.com
ihaveabadreputation.com

Search

Clear

Export CSV

URL	Category	Reputation
cisco.com	Computer and Internet Info, Business and Economy	<i>Well known</i>
developmentserver.com	Computer and Internet Info	<i>Benign sites</i>
google.com	Search Engines	<i>Well known</i>
ihaveabadreputation.com	Malware Sites	<i>High risk</i>



ISE remediation in using pxGrid

Edit Instance

Instance Name SampleResponses
Module pxGrid Mitigation(v1.0)

Description

Enable Logging ☒ On ☐ Off

Save

Cancel

Configured Remediations

Remediation Name	Remediation Type	Description
------------------	------------------	-------------

No configured remediations available

Add a new remediation of type

Mitigate Destination

Add

Edit Remediation

Remediation Name

Remediation Type

Mitigate Destination

Description

Mitigation Action

quarantine

quarantine
unquarantine
shutdown

Whitelist
(an optional list of networks)

Create

Cancel

Edit Instance

Instance Name SampleResponses
Module pxGrid Mitigation(v1.0)

Description

Enable Logging ☒ On ☐ Off

Save

Cancel

Configured Remediations

Remediation Name	Remediation Type	Description
QuarantineDestination	Mitigate Destination	
QuarantineSource	Mitigate Source	

Add a new remediation of type

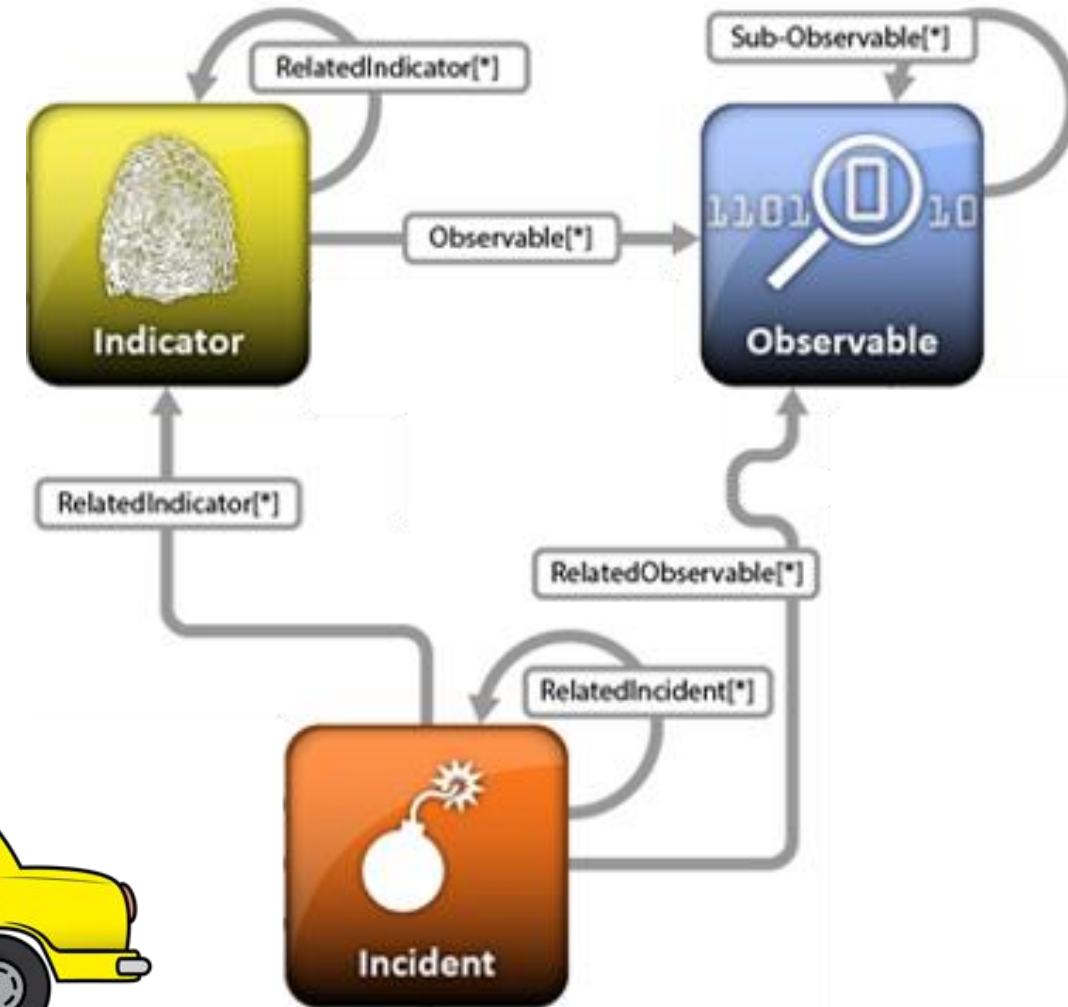
Mitigate Destination

Add

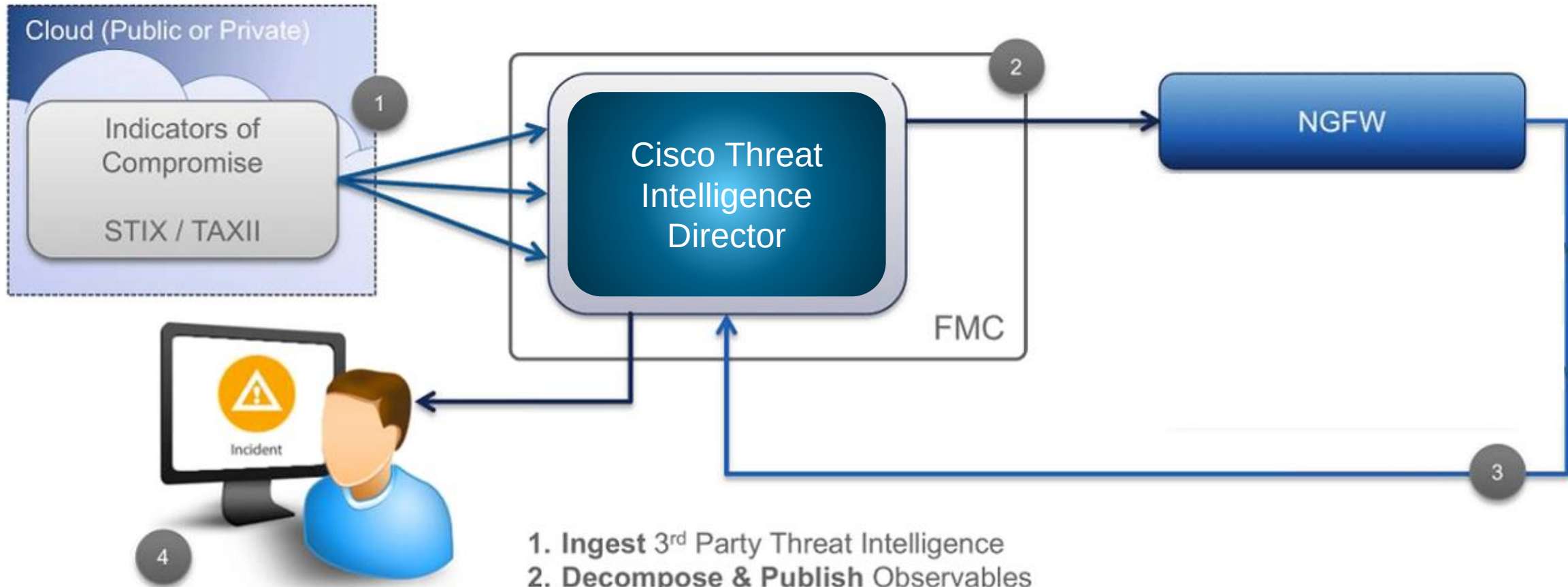


Cisco Threat Intelligence Director (CTID)

- Uses customer threat intelligence to identify threats
- Automatically blocks supported indicators on Cisco NGFW
- Provides a single integration point for all STIX and CSV intelligence sources



Cisco Threat Intelligence Director Overview



Hail a TAXII !!

- Free source of TAXII feeds
- Website URL: <http://hailataxii.com>
- Multiple feeds
- To configure the TAXII intelligence source
 - URL: <http://hailataxii.com/taxii-discovery-service>
 - USERNAME: **guest**
 - PASSWORD: **guest**



Poll to Start.

WHAT IS IT?

Hail a TAXII.com is a repository of Open Source Cyber Threat Intelligence feeds in STIX format. There are currently 727165 indicators, last updated Wed Jan 11 16:25:00 2017 UTC.

AVAILABLE FEEDS

guest.Abuse_ch
guest.CyberCrime_Tracker
guest.EmergingThreats_rules
guest.Lehigh_edu
guest.MalwareDomainList_Hostlist
guest.blutmagie_de_torExits
guest.dataForLast_7daysOnly
guest.dshield_BlockList
guest.phishtank_com

HOW TO CONNECT

Our data is accessible via the TAXII-HTTP Message Protocol. (1.0 & 1.1)
The discovery service is located at <http://hailataxii.com/taxii-discovery-service>

Anonymous connections are accepted.
Clients that require login details can use HTTP-Basic user=guest, password=guest.

CONTACT US

For questions or comments, please contact us using info@hailataxii.com.

Deployment Designs Use Case



Use Case

Internet Edge Firewall

Requirement

Connectivity and Availability Requirement:

- High Availability ROUTED mode
- Firewall should support Router or Transparent Mode

Routing Requirements:

- Static and BGP Routing
- Dynamic NAT/PAT and Static NAT

Security Requirements:

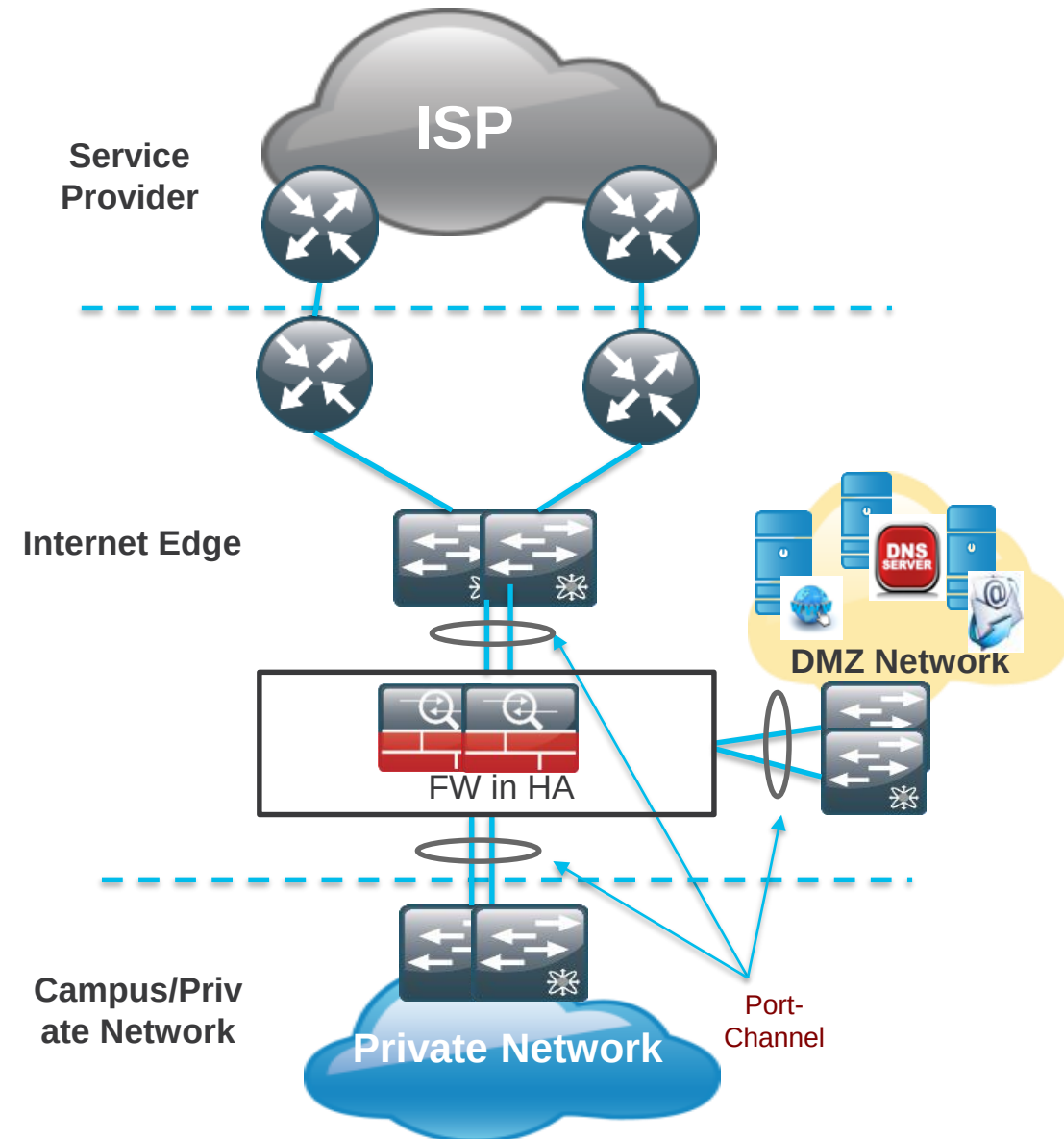
- Application Control + URL Acceptable Use enforcement
- IPS and Malware protection
- SSL Decryption

Authentication Requirements:

- User authentication and device identity

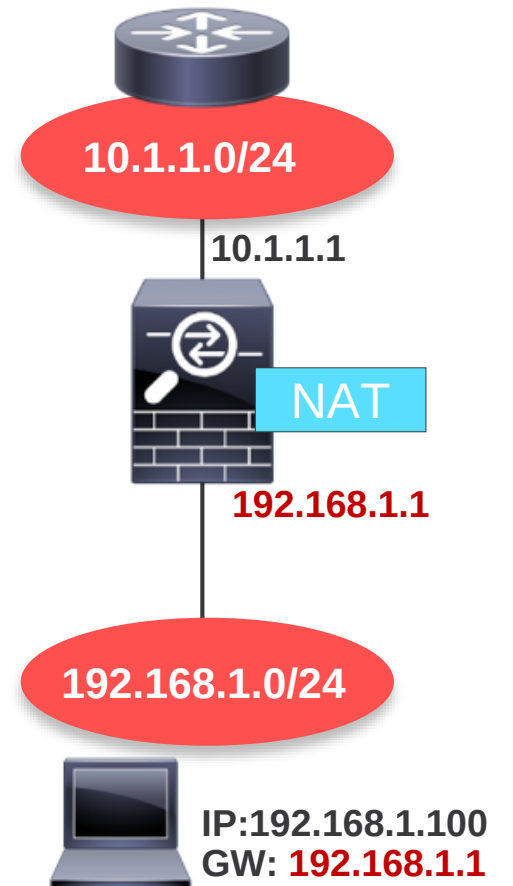
Solution

Security Application: Firepower Threat Defense application with FMC



Firewall Design: Modes of Operation

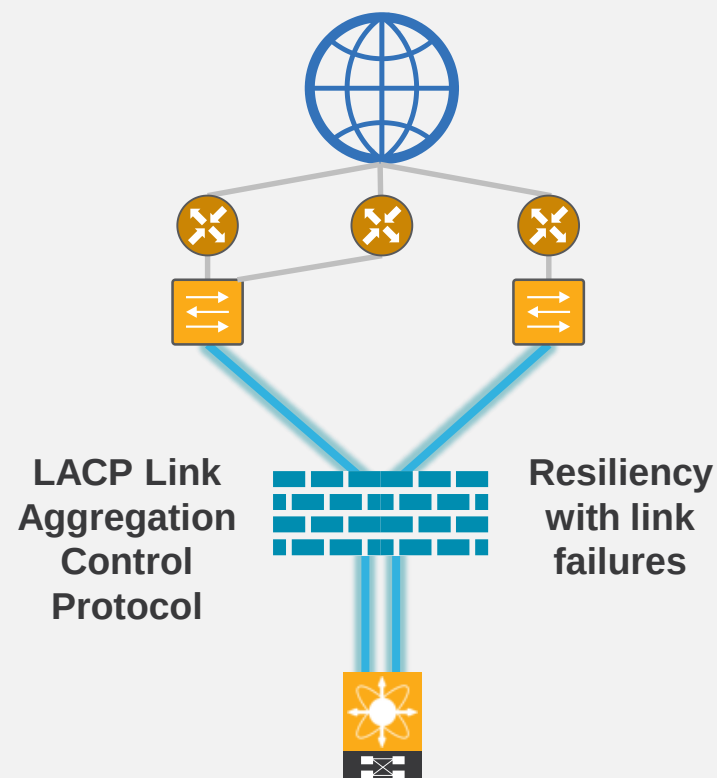
- **Routed Mode** is the traditional mode of the firewall. Two or more interfaces that separate L3 domains – Firewall is the Router and Gateway for local hosts.
- **Transparent Mode** is where the firewall acts as a bridge functioning at L2.
 - Transparent mode firewall offers some unique benefits in the DC.
 - Transparent deployment is tightly integrated with our ‘best practice’ data center designs.



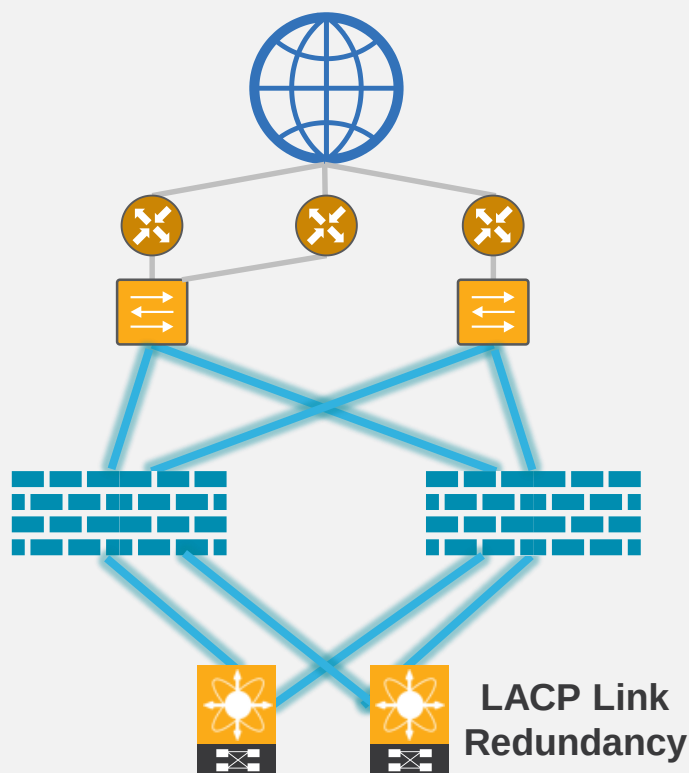
Link and Platform Redundancy Capabilities

Firewall Link Aggregation – High Availability - Clustering

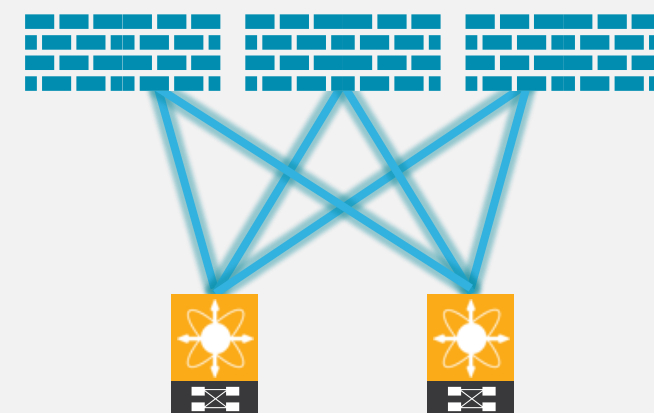
Link Redundancy



Active / Standby HA



Inter-chassis Clustering



Combine up to

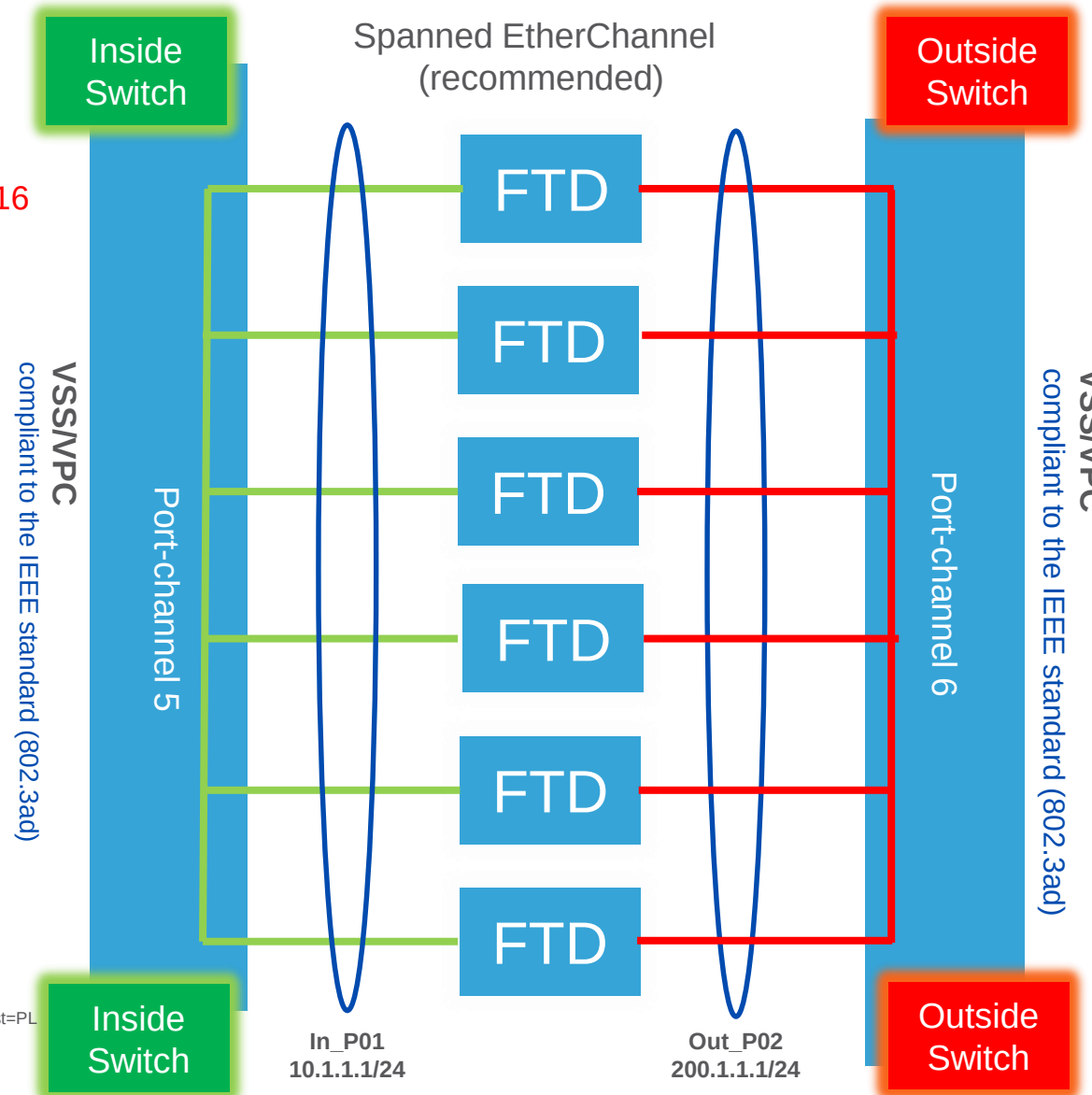
6

9300 blades or
4100 appliances

Clustering

Pay-As-You-Grow

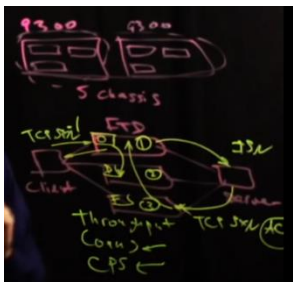
- Traditional ASA 16 node cluster
- FTD 6 nodes today will scale to 16 in the near future



Benefits

- High Scale: NGFW
- Network Integration: Routing, switching, inter-site DC extensions
- High Density: 40G/100G
- Clustering: Intra-chassis, Inter-chassis, Inter-site
- Consistent Policy Management

Note: L3 PBR and ECMP models are supported



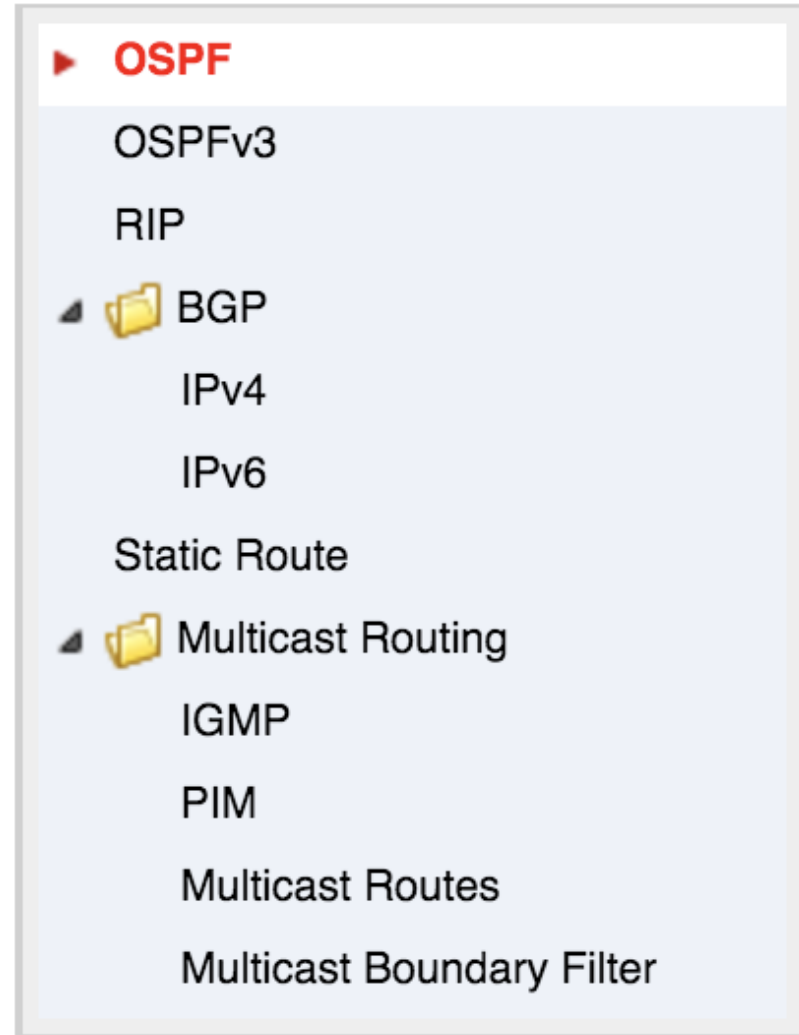
Cisco Security Chalk Talk - NGFW Clustering Technology
https://www.youtube.com/watch?v=yt8Cc4tS0kE&t=38s&index=3&list=PLFT-9JpKjRTANXKBmLbQ611TPYLxbUL_0

Routing Requirements



Routing Protocol support

- OSPF and OSPFv3 (IPv6)
- BGP (IPv4 & IPv6)
- Static Route
 - Tunneled Route support for VPNs
 - Reverse Route Injection for VPNs
- Multicast Routing
 - IGMP
 - PIM
- EIGRP via FlexConfig





#	Directi...	Type	Source Interface Objects	Destination Interface Objects	Original Packet			Translated Packet			Options	
					Original Sources	Original Destinations	Or... Se...	Translated Sources	Translated Destinations	Tr... Se...		
▼ NAT Rules Before												
1		Static	 any	 any	 IPv4-Private-192	 IPv4-Private-192		 IPv4-Private-192	 IPv4-Private-192		 Dns:false  no-proxy-arp	 
▼ Auto NAT Rules												
#		Dynamic	 FTD-Guest-Zone	 FTD-Outside-Zone	 Home-Net			 Interface			 Dns:false	 
▼ NAT Rules After												



Rate limiting Cloud File Sharing Traffic

- QOS Policy is a new policy type with separate policy table
- Not associated with an Access Control Policy – directly associated with devices

Add Rule



Name:* ☒ Enabled

Apply QoS On: ▼

Traffic Limit Per Interface

Download Limit: Mbits/sec (Range: 0.008-100000 Mbits/sec)

Upload Limit: Mbits/sec (Range: 0.008-100000 Mbits/sec)

Interface Objects

Networks

Users

Applications

Ports

URLs

SGT/ISE Attributes

Comments

Application Filters

Clear All Filters

Available Applications (9)

Selected Applications and Filters (9)

Search by name

amazon

User-Created Filters

Risks (Any Selected)

	Very Low	1097
	Low	806
	Medium	1000
	High	212
	Very High	122

amazon

☐ All apps matching the filter

- ☐ Amazon
- ☐ Amazon Ads System
- ☒ Amazon Cloud Drive
- ☒ Amazon Cloud Drive Authenticate
- ☒ Amazon Cloud Drive Download

Applications

- ☐ Amazon
- ☐ Amazon Ads System
- ☒ Amazon Cloud Drive
- ☒ Amazon Cloud Drive Authenticate
- ☒ Amazon Cloud Drive Download
- ☒ Amazon Cloud Drive Upload

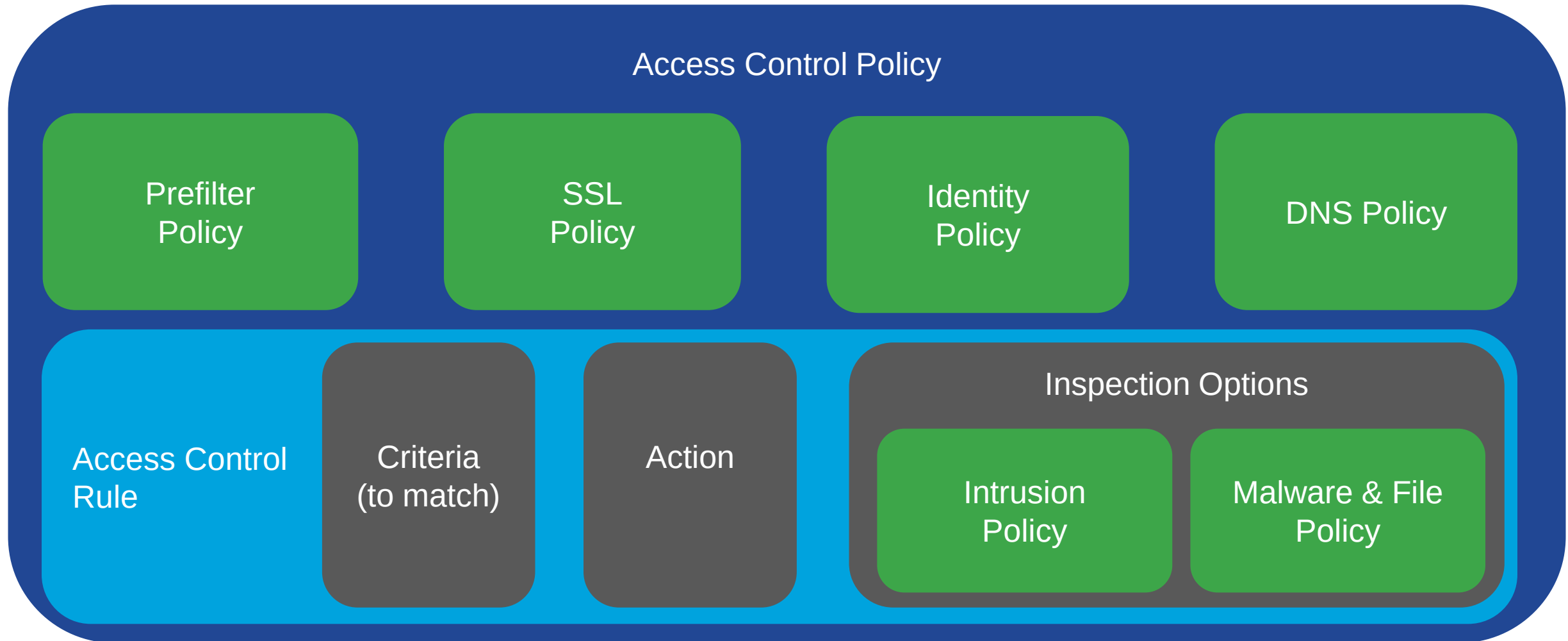
Add to Rule

Security Requirements



Access Control Policy

The glue that ties everything together





Access Control Policy blocking inappropriate content

Overview Analysis **Policies** Devices Objects AMP Deploy System Help jfanelli

Access Control ▶ Access Control Network Discovery Application Detectors Correlation Actions

Home Gateway FTD Policy

Enter a description

Save Cancel

Identity Policy: None

SSL Policy: None

Inheritance Settings | Policy Assignments (1)

Rules Security Intelligence HTTP Responses Advanced

Filter by Device Add Category Add Rule Search Rules

#	Name	Source Zones	D... Z...	S... N...	D... N...	V...	U...	A...	Sr...	De	URLs	Action				
▼ Mandatory - Home Gateway FTD Policy (1-1)																
1	Block Bad Stuff	FTD-Inside-Zone	any	any	any	any	any	any	any	any	Gambling (Any Reputation) Malware Sites (Any Reputation) Marijuana (Any Reputation) SPAM URLs (Any Reputation) Adult and Pornography (Any Reputation)	Interactive Block				0
▼ Default - Home Gateway FTD Policy (2-2)																
2	Permit The Rest	FTD-Inside-Zone	any	any	any	any	any	any	any	any	any	Allow				0
Default Action											Intrusion Prevention: Security Over Connectivity					

Security Intelligence & DNS Global Settings

Whitelist / Blacklist capabilities

Identity Policy: [Main Identity Policy](#)

SSL Policy: [SSL Decryption Policy](#)

[Inheritance Settings](#) | [Policy Assignments \(1\)](#)

Rules **Security Intelligence** HTTP Responses Advanced

Available Objects **Available Zones**

Search for a Network

Networks **URLs**

Global-Blacklist
Global-Whitelist
Attackers
Bogon
Bots
CnC
Dga
Exploitkit
Malware
Open_proxy
Open_relay
Phishing
Response

Any
CVO-Zone
DMZ-Zone
FTD-Inside-Zone
FTD-Outside-Zone
Inside-Zone
Outside-Zone

Add to Whitelist
Add to Blacklist

DNS Policy Default DNS Policy

Whitelist (2)

Networks

Global Whitelist (Any Zone)

URLs

Global Whitelist for URL (Any Zone)

Blacklist (2)

Networks

Global Blacklist (Any Zone)

URLs

Global Blacklist for URL (Any Zone)

Granular SSL Decrypt

Can specify by application, certificate fields / status, ciphers, etc.

Overview Analysis **Policies** Devices Objects AMP Deploy System Help jfanelli

Access Control ▸ SSL Network Discovery Application Detectors Correlation Actions ▾

SSL Decryption Policy

Save Cancel

Add Rule

Name: Outbound FTD Decrypt ☒ Enabled Insert: below rule 1

Action: Do not decrypt with: issuer-ca-signer ☐ Replace Key

Zone: Users Applications Ports Category Certificate **CN** Cert Status Cipher Suite Version Logging

Available:

- Do not decrypt
- Block
- Block with reset
- Monitor
- CN=*.logmein.com
- CN=*.mozilla.org
- CN=*.rhn.redhat.com
- CN=*.sls.microsoft.com
- CN=*.update.microsoft.com
- CN=*.update.microsoft.com
- CN=account.live.com
- mail2web.com

Subject DN(s) (1): mail2web.com

Issuer DN(s) (0): any

Add to Subject Add to Issuer

Enter DN or CN Add Enter DN or CN Add

Custom IPS Policy

[Overview](#) [Analysis](#) **[Policies](#)** [Devices](#) [Objects](#) [AMP](#)

[Access Control ▶ Intrusion](#) [Network Discovery](#) [Application Detectors](#) [Correlation](#) [Actions ▼](#)

[Deploy](#) [System](#) [Help ▼](#) [jfanelli ▼](#)

Edit Policy: Custom Intrusion Policy

Policy Information

Rules

Firepower Recommendations

Advanced Settings

Global Rule Thresholding

Policy Layers

My Changes

Rules

Security Over Connectivity

Rules

Global Rule Thresholding

Policy Information [< Back](#)

Name

Custom Intrusion Policy

Description

Aggressive IPS

Drop when Inline

☒

Base Policy

Security Over Connectivity

[Manage Base Policy](#)

This policy has 10508 enabled rules

→ 243 rules generate events

✗ 10265 rules drop and generate events

[Manage Rules](#)

[View](#)

[View](#)

No recommendations have been generated. [Click here to set up Firepower recommendations.](#)

Commit Changes

Discard Changes

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Intrusion Policy and Network Discovery Policy

- Firepower Recommended Rules automatically tunes your Snort rules for the applications, servers, and hosts on your network

The screenshot shows the Cisco Firepower Management Center interface. At the top, there is a navigation bar with tabs: 'on', 'Network Discovery', 'Application Detectors', 'Correlation', and 'Actions'. Below this, the 'Network Discovery' tab is active, displaying the 'bango IPS Policy' configuration. A blue header bar reads 'Firepower Recommended Rules Configuration'. Below this, a green globe icon is followed by the text 'Firepower changed 16052 rule states for 64 hosts'. A list of actions follows: a green arrow pointing right for 'Set 103 rules to generate events', a red 'X' for 'Set 7038 rules to drop and generate events', and a green arrow pointing right for 'Set 8911 rules to disabled'. At the bottom, it states 'Policy is using the recommendations. Click to change recommendations' and 'Last generated: 2017 Jan 7 12:26:37'.

on Network Discovery Application Detectors Correlation Actions ▼

bango IPS Policy

Firepower Recommended Rules Configuration

 **Firepower changed 16052 rule states for 64 hosts**

- ➔ Set 103 rules to generate events
- ✗ Set 7038 rules to drop and generate events
- ➔ Set 8911 rules to disabled

Policy is using the recommendations. Click to change recommendations

Last generated: 2017 Jan 7 12:26:37

FirePower NGFW Intrusion events

	Time	Priority	Impact	Inline Result	Source IP	Source Country	Destination IP	Destination Country	Message	Classification
↓	2015-11-09 20:24:36	medium	1	↓	172.16.0.75		54.2			Informa
↓	2015-11-09 20:12:48	medium	1	↓	172.16.0.75		74.1			Informa
↓	2015-11-09 20:25:59	high	2	↓	10.120.10.123		10.1			corporate
↓	2015-11-09 20:25:59	high	2	↓	10.0.69.125		10.1			corporate
↓	2015-11-09 20:25:59	high	2	↓	10.131.12.20		10.0			corporate
↓	2015-11-09 20:25:59	high	2	↓	10.0.68.103		172.			corporate
↓	2015-11-09 20:25:59	high	2	↓	10.0.37.123		10.1			corporate
↓	2015-11-09 20:25:59	high	2	↓	10.0.37.123		10.0			corporate
↓	2015-11-09 20:25:59	high	2	↓	10.120.10.123		10.1			corporate
↓	2015-11-09 20:25:59	high	2	↓	10.120.10.123		10.1			corporate
↓	2015-11-09 20:25:59	high	2	↓	10.131.12.20		10.0			corporate
↓	2015-11-09 20:25:59	high	2	↓	10.0.95.115		10.0			corporate
↓	2015-11-09 20:25:59	high	2	↓	10.0.69.125		10.0			corporate
↓	2015-11-09 20:25:12	high	2	↓	172.16.0.100		10.1			corporate
↓	2015-11-09 20:25:12	high	2	↓	10.0.228.99		10.1			corporate
↓	2015-11-09 20:25:12	high	2	↓	172.16.0.100		172.			corporate

Impact Flag	Administrator Action	Why
 1	Act immediately, vulnerable	Event corresponds to vulnerability mapped to host
 2	Investigate, potentially vulnerable	Relevant port open or protocol in use, but no vuln mapped
 3	Good to know, currently not vulnerable	Relevant port not open or protocol not in use
 4	Good to know, unknown target	Monitored network, but unknown host
 0	Good to know, unknown network	Unmonitored network



Malware and File Analysis

Attached to Access Policy

Overview

Analysis

Policies

Dev

Access Control ▶ Malware & File

Home File Policy

Enter a description

Rules

Advanced

Edit File Rule

Application Protocol

Any

Direction of Transfer

Any

Action

Block Malware

☐ Spero Analysis for MSEXE

☒ Dynamic Analysis

☒ Capacity Handling

☐ Local Malware Analysis

☒ Reset Connection

Store Files

☒ Malware

☐ Unknown

☐ Clean

☐ Custom

File Type Categories

Office Documents

16

Archive

17

Multimedia

2

File Types

Search name and description

7Z (7-Zip compressed file)

ACCDB (Microsoft Access 2003 file)

Selected File Categories and Types

Category: Executables

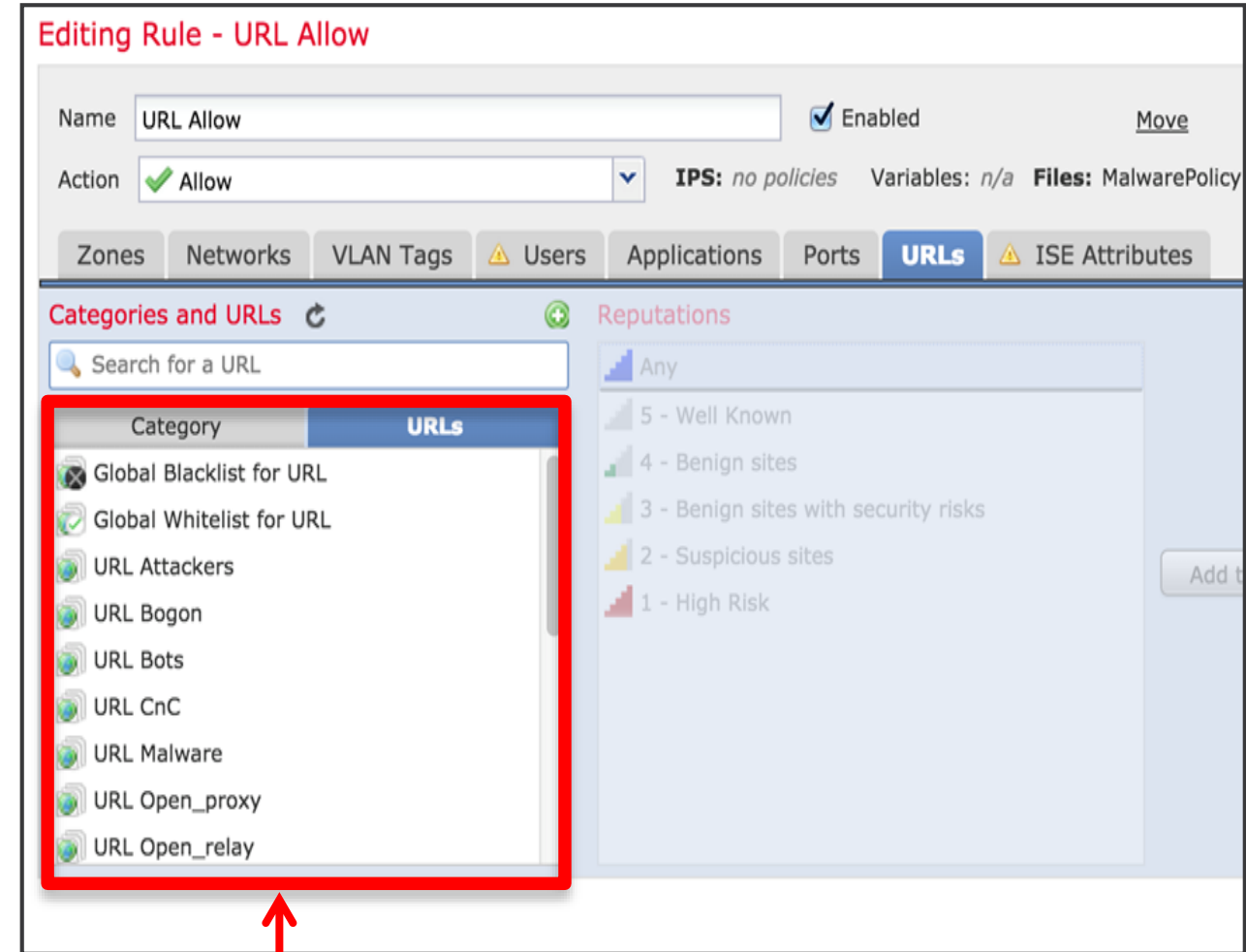
Category: Office Documents

File Types	Application Protocol	Direction	Action
Category: Executables Category: Office Documents	Any	Any	Block Malware with Reset Dynamic Analysis Capacity Handling Store files of disposition: Malw
Category: Local Malware Analysis Capable Category: Dynamic Analysis Capable Category: System files Category: Graphics (6 more...)	Any	Any	Detect Files



URL-Based Security Intelligence

- Extension of IP-based SI
- TALOS dynamic feed, 3rd party feeds and lists
- Multiple categories: Malware, Phishing, CnC,...
- Multiple Actions: Allow, Monitor, Block, Interactive Block,...
- Policy configured via Access Rules or blacklist
- IoC tags for CnC and Malware URLs
- New Dashboard widget for URL SI
- Black/White-list URL with one click



**URL-SI
Categories**

DNS Inspection

- Security Intelligence support for domains
- Addresses challenges with fast-flux domains
- Cisco provided and user defined DNS lists: CnC, Spam, Malware, Phishing
- Multiple Actions: Block, Domain Not Found, Sinkhole, Monitor
- Indications of Compromise extended with DNS Security Intelligence

Rules			
#	Name	DNS Lists	Action
Whitelist			
1	Global DNS Whitelist	Global Whitelist for DNS	Whitelist
Blacklist			
2	Global DNS Blacklist	Global Blacklist for DNS	Domain Not Found
3	DNS Drop	DNS Spam DNS_DROP	Drop
4	DNS Monitor	DNS Bots DNS_Tor_exit_node DNS_MONITOR	Monitor
5	DNS Nxdomain	DNS Open_proxy DNS_NXDOMAIN	Domain Not Found
6	DNS Sinkhole	DNS Attackers DNS CnC DNS Malware DNS Phishing (2 more...)	Sinkhole

DNS List

Action



Identity Policy based on Passive Authentication

Overview

Analysis

Policies

Devices

Objects

AMP

Deploy

1 System

Help ▾

jfanelli ▾

Access Control ▸ Identity

Network Discovery

Application Detectors

Correlation

Actions ▾

Main Identity Policy

Save Cancel

Enter a description

Rules

Active Authentication

Add Category Add Rule

#	Name	Source Zones	Dest Zones	Sou... Net...	Dest Ne...	VL...	Src...	De...	Re...	Action	Auth Type
---	------	--------------	------------	---------------	------------	-------	--------	-------	-------	--------	-----------

Administrator Rules

This category is empty

Standard Rules

1	FTD Internet Auth	FTD-Inside-Zone (Routed)	FTD-Outside-Zone (Routed)	any	any	any	any	any	home	Passive Authenticatio	none	
2	Internet Auth	Inside-Zone (ASA)	Outside-Zone (ASA)	any	any	any	any	any	home	Passive Authenticatio	none	

Root Rules

Zones

Networks

VLAN Tags

Ports

Realm *

☐ Use active authentication if passive authentication cannot identify user

* Required Field



Access Control Policy Identity Control

Can Mix and Match AD & ISE Identity Groups (Guest, BYOD, etc.)

Editing Rule - Block Bad Stuff

Name: ☒ Enabled [Move](#)

Action: ❌ Interactive Block IPS: no policies Variables: n/a Files: no inspection Logging: no logging

Zones Networks VLAN Tags **Users** Applications Ports **URLs** ISE Attributes Inspection **Logging** Comments

Available Realms ↻

Search by name or value

- Special Identities
- home

Available Users ↻

Search by name or value

- home/*
- Domain Admins
- Domain Users**
- administrator
- defaultaccount
- emilia
- jfanelli
- kim
- krbtgt
- sofia

[Add to Rule](#)

Selected Users (2)

- Special Identities/Guest
- home/Domain Users

[OK](#) [Cancel](#)



ISE Integration



- pxGrid feed to retrieve from ISE:
 - AD Username (Group lookup via AD Realm)
 - Device type profile & location
 - TrustSec Security Group Tag (SGT)
- Ability to exert control based on the above in rules
 - i.e. block HR users from using personal iPads
- Reduces ACL size and complexity



TrustSec Security Group Tag based identity from ISE

Can also reference Identity Services Engine identified Device Profiles

Editing Rule - Block Bad Stuff

Name: ☒ Enabled [Move](#)

Action: ✖ Interactive Block IPS: no policies Variables: n/a Files: no inspection Logging: no logging

Zones Networks VLAN Tags **Users** Applications Ports **URLs** **ISE Attributes** Inspection **Logging** Comments

Available ISE Session Attributes

Search by name or value

- Security Group Tag
- Device type
- Location IP

Available ISE Metadata

Search by name or value

- ANY
- SGT_Auditor
- SGT_BYOD
- SGT_Contractor**
- SGT_Developer
- SGT_DevelopmentServers
- SGT_Employee
- SGT_Guest
- SGT_NetworkServices
- SGT_PCIServers

[Add to Rule](#)

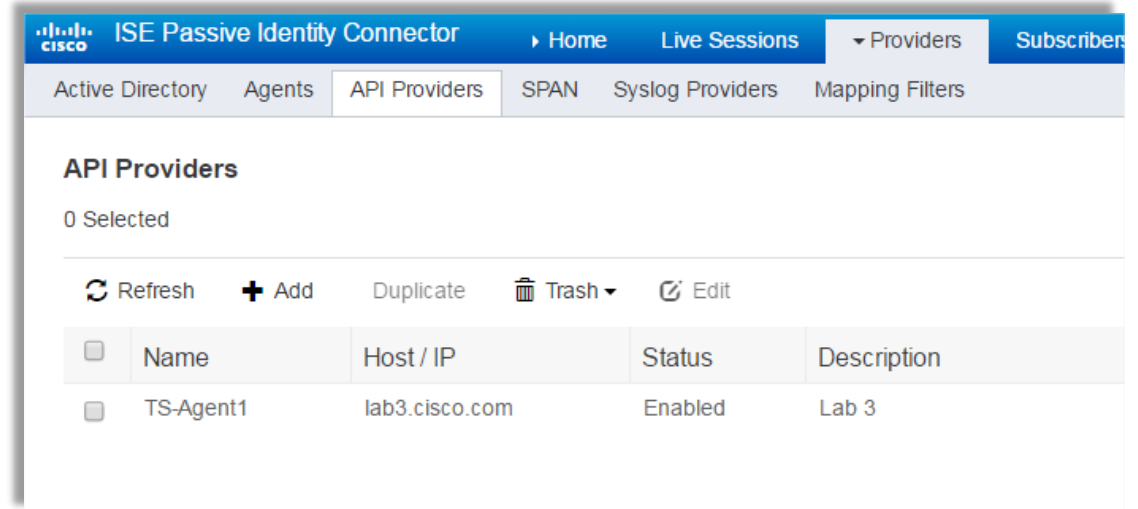
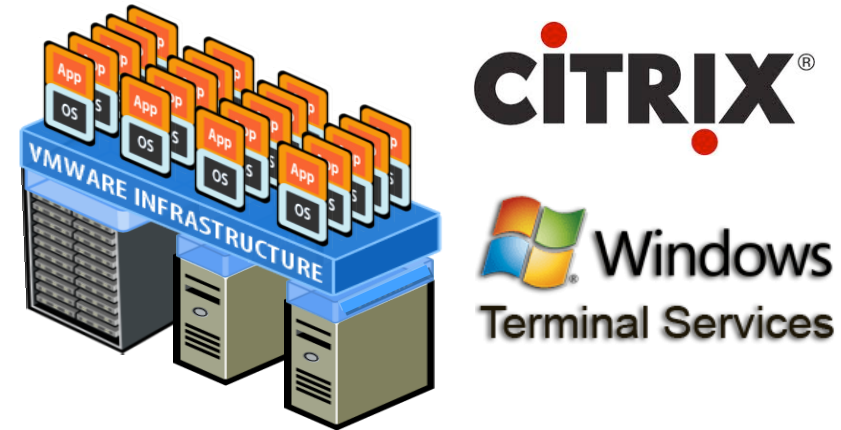
Selected Source ISE Metadata (3)

- SGT_Employee
- SGT_Guest
- SGT_Contractor

[Add](#)

REST API Provider

- Designed for use with Terminal Services Agent
- Can also be used by custom integrations
- Uses certificate-based authentication
- User information is sent to the passive ID node over SSL in JSON format



VDI Environments

Enables you to interface with network applications such as the TS-Agent on a Citrix server, where all users have the same IP address but are assigned unique ports.

▶ Home

Live Sessions

▶ Providers

Subscribers

▶ Certificates

Troubleshoot

Reports

▶ Administration

▶ Settings

License

Refresh

Every 1 minute

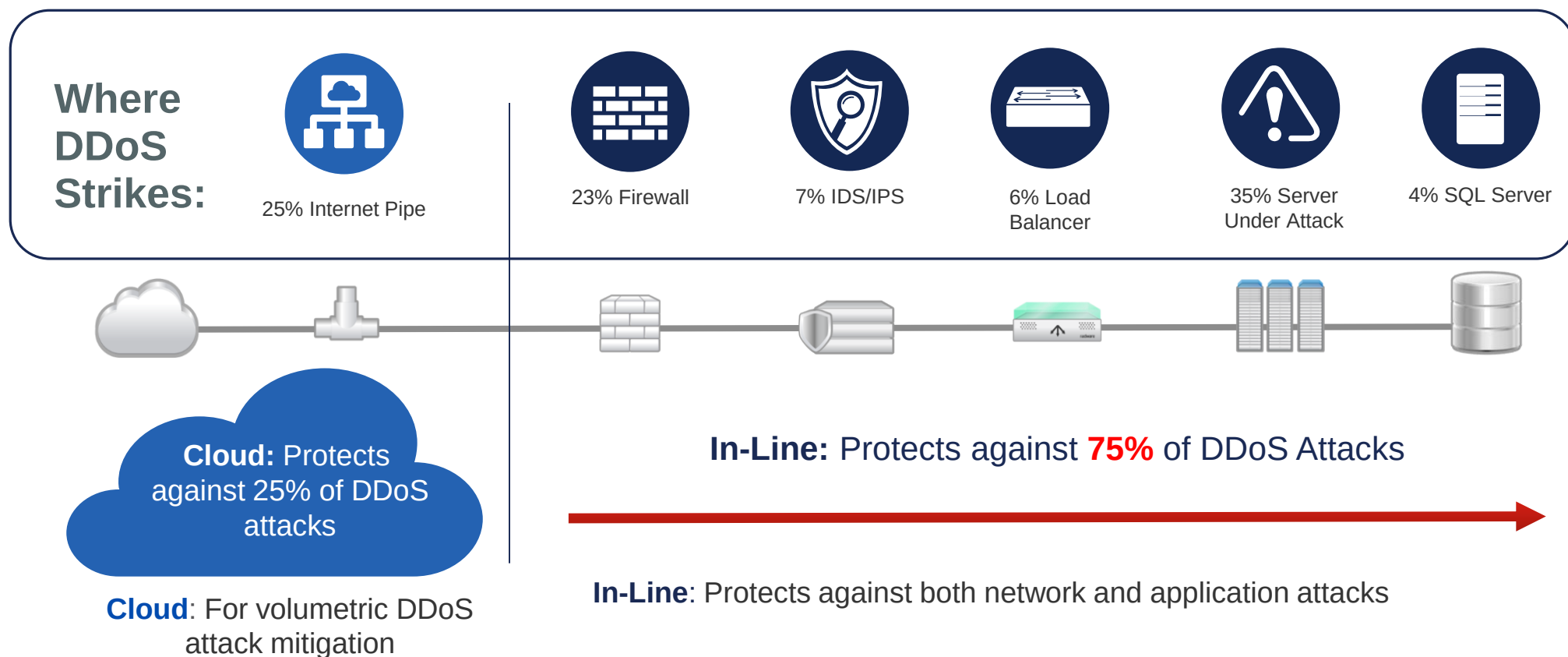
Show

Latest 20

Endpoint ID	Identity	IP Address	Provider	Source Port Start	Source Port End	Source First Port	Ts Agent Id
Endpoint ID	Identity	IP Address		Source Port Start	Source Port End	Source First Port	Ts Agent Id
13.13.13.13-100-199	frank	10.56.15.13	Rest	100	199	100	agent2
12.12.12.12-300-399	jack	10.56.15.12	Rest	300	399	100	agent1
12.12.12.12-200-299	sam	10.56.15.12	Rest	200	299	100	agent1
12.12.12.12-100-199	john	10.56.15.12	Rest	100	199	100	agent1

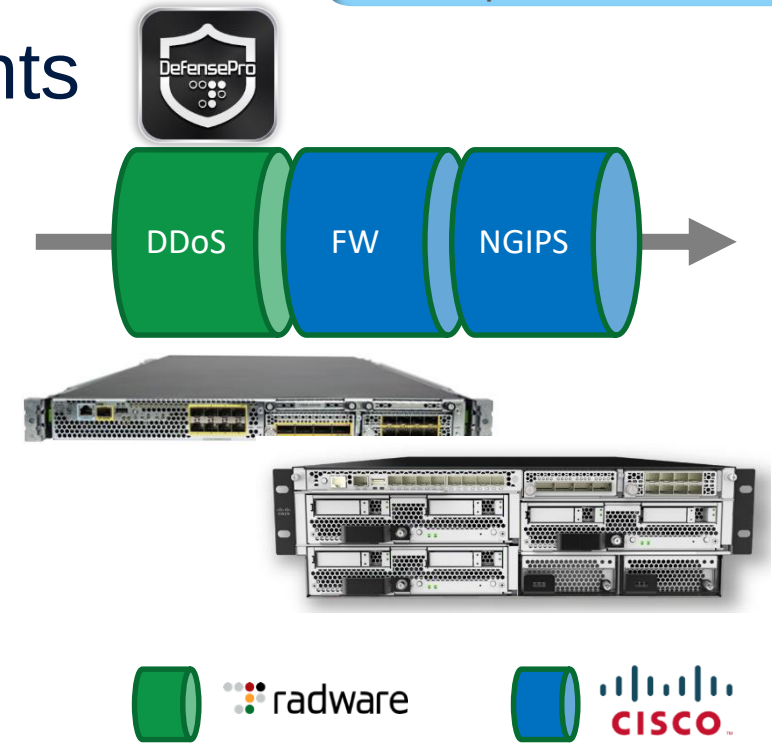
Radware and Denial of Service Attack Landscape

DDoS Attack Surface – Hybrid mitigation strategy

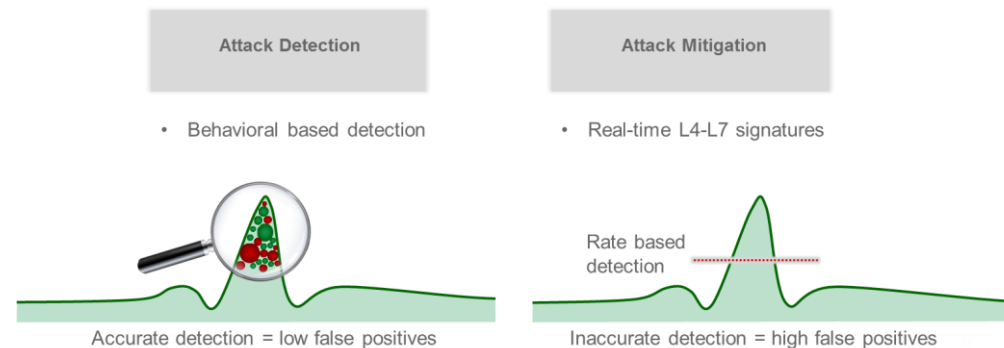


Firepower DDoS Solution Components

- Cisco Firepower is a scalable, carrier & enterprise -grade, multi-service security appliance featuring:
 - Radware DDoS Decorator App (OEM)
 - Cisco ASA firewall
 - Cisco NGIPS (Sourcefire – Threat Defense)
- What is required?
 - Firepower Chassis (FXOS 1.1.4+)
 - DDoS License (Virtual DefensePro)
 - Vision Management Software
 - **Cloud DDoS** *CSCO FY18 Q1 (Oct 15, 2017)
 - Hybrid, Always on & On Demand



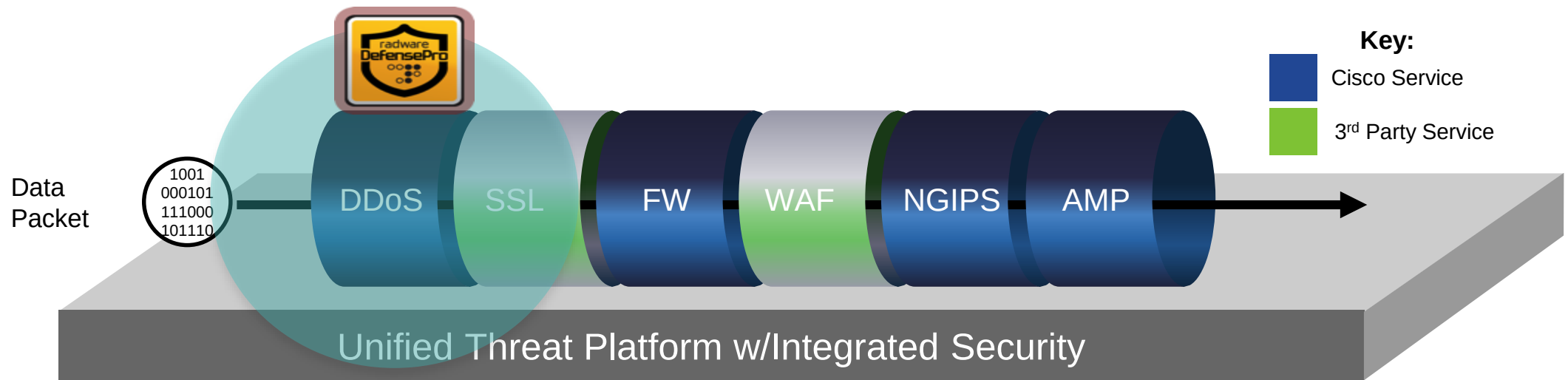
Behavioral Based Detection and Mitigation





The Firepower 4100/9300 Transforms Security Service Integration

- Radware vDP is our first 3rd Party component of the new Architecture
- We are adding DDoS Application Services to the ingress interfaces of the Firepower 4100/9300



Limited effectiveness

Maximum protection

Increased latency

Highly efficient

Slows network

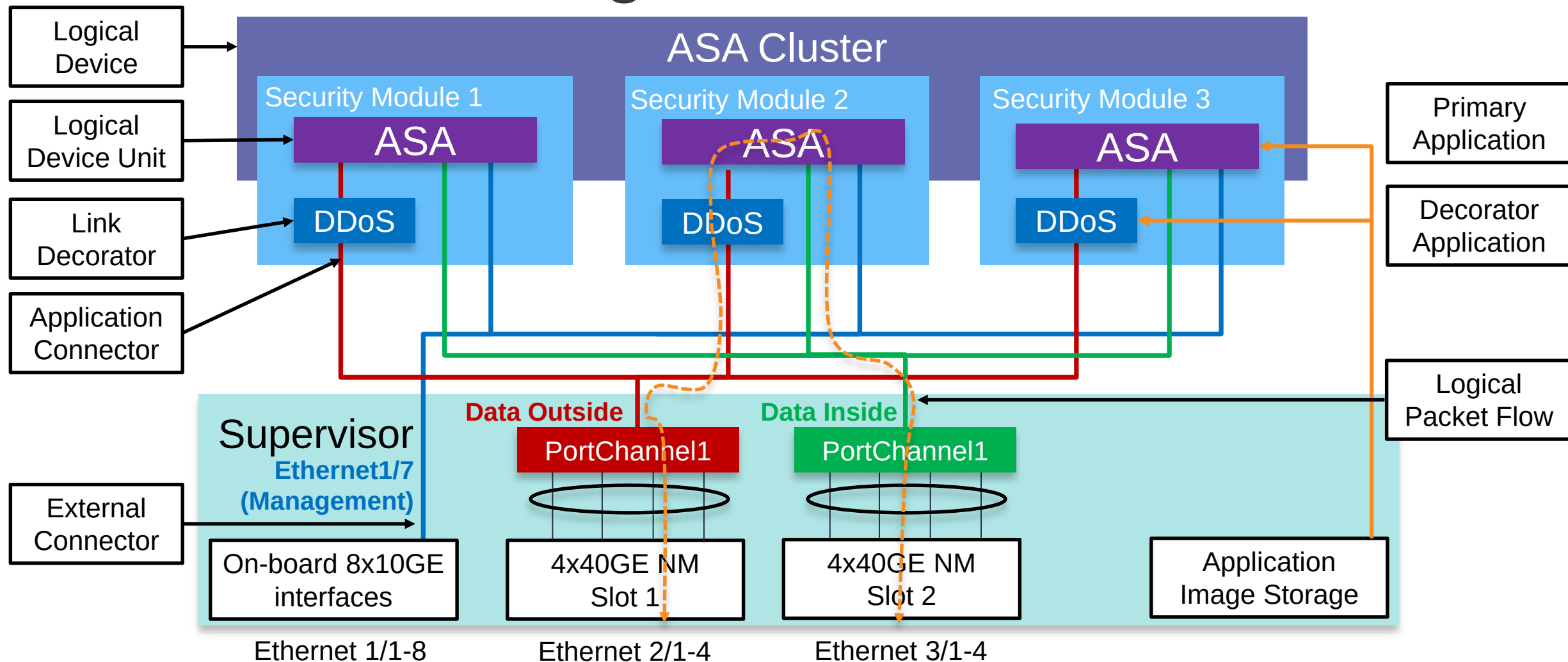
Scalable processing

Static & Manual

Dynamic



Security Services Architecture with DDoS running



Platform and License Support

Firepower	ASA	FTD
9300 – SM 44	✓	✓
9300 – SM 36	✓	✓
9300 – SM 24	✓	✓
4150	✓	✓
4140	✓	✓
4120	✓	✓
4110	✗ * QA	✓

vDP License Support:
FPR-RVDP-10G
FPR-RVDP-5G
FPR-RVDP-2G
FPR-RVDP-1G
FPR-RVDP-500M
FPR-RVDP-200M
SWSS 12-60 Months

Abbreviation Key!



ASA = Adaptive Security Appliance

FTD = Firepower Threat Defense

FPS = Firepower Services

FMC = Firepower Management Center

FDM = Firepower Device Manager

NGFW = Next Generation Firewall

NGIPS = Next Generation Intrusion
Prevention System

AMP = Advanced Malware Protection

API = Application Programming Interface

ISE = Identity Services Engine

IoC = Indicator of Compromise

PAN = Place to cook your eggs

Useful links

FTD: Common Practices Guide:

http://cisco.lookbookhq.com/ngfw_ftd_common-practices/ftd-common-practices

Short how-to videos:

<https://www.youtube.com/channel/UCwnm1oSSz8pPwDyFzFS5k3w/playlists>

Lab minutes videos: www.labminutes.com

BU videos:

https://www.youtube.com/channel/UCxTz5VApACLnH5_SDjtfoNg/videos?view_as=subscriber

Thank you.

